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Research

AI at 70: 14 lessons from a lifetime of boom and bust

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Artificial intelligence marks 70 years since Dartmouth Summer Research Project

"Context is all you need": clues from the past for what comes next

Seventy years ago today, the curtain came down on the summer workshop that gave birth to artificial intelligence. Dwight Eisenhower was in the White House, the Suez Crisis was gripping Britain and Elvis Presley topped the US charts with "Don't Be Cruel" and "Hound Dog".

The Dartmouth Summer Research project brought together a dozen or so academics convinced that every aspect of intelligence could in principle be described precisely enough for a machine to simulate it. Emboldened by the techno-optimism of the age, John McCarthy, who coined the name AI, wrote that "we think that a significant advance can be made… if a carefully selected group of scientists work on it together for a summer."

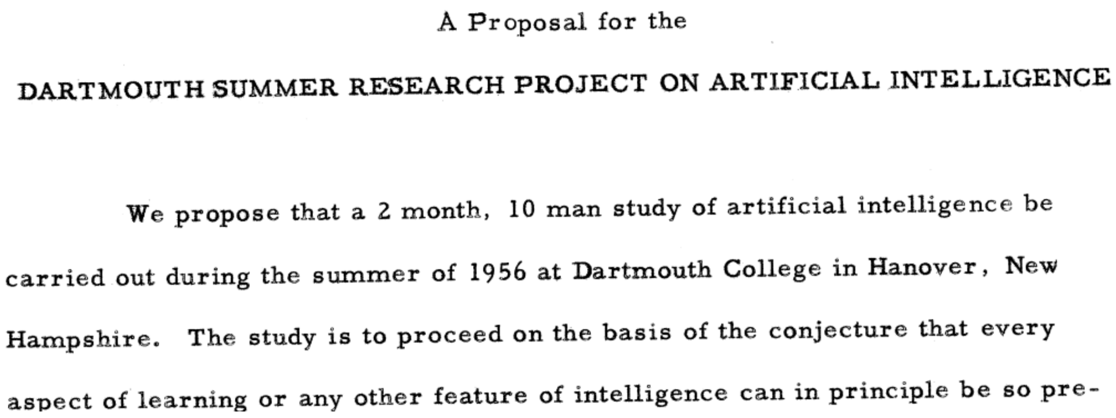
Since then, AI has lived a lifetime of summers and winters, punctuated by stunning breakthroughs and false dawns. As it is now the cornerstone of a new investment and valuation boom, the past 70 years are a treasure trove of clues for what to expect in the future.

To paraphrase the research paper that sparked the current boom in generative AI in 2017, "Attention is all you need", you could say "context is all you need" to understand what is to come. The signals are there, from non-linear growth to infrastructure constraints and valuation booms and busts.

Some will say "this time is different". Maybe. But the following charts give a flavour of what could happen if it isn't. To quote the UK No. 1 single from 70 years ago, sung by Doris Day: "Whatever Will Be, Will Be (Que Sera, Sera)".

14 lessons from a lifetime of boom and bust

1. AI progress is non-linear
2. Progress exceeds Moore's law
3. Today's leading tech may not be tomorrow's
4. The pipeline matters
5. Cheaper AI will likely boost spending
6. Software revolutions need hardware
7. Hardware is often a bottleneck
8. Globalisation has made tech cheaper, and more at risk
9. Tech revolutions take time to pay
10. Consumers have adopted AI faster than businesses
11. The biggest valuation booms are often tech booms
12. This time is different – and will need to be
13. Early leads do not guarantee long-term winners
14. Long-term trends tend to persist



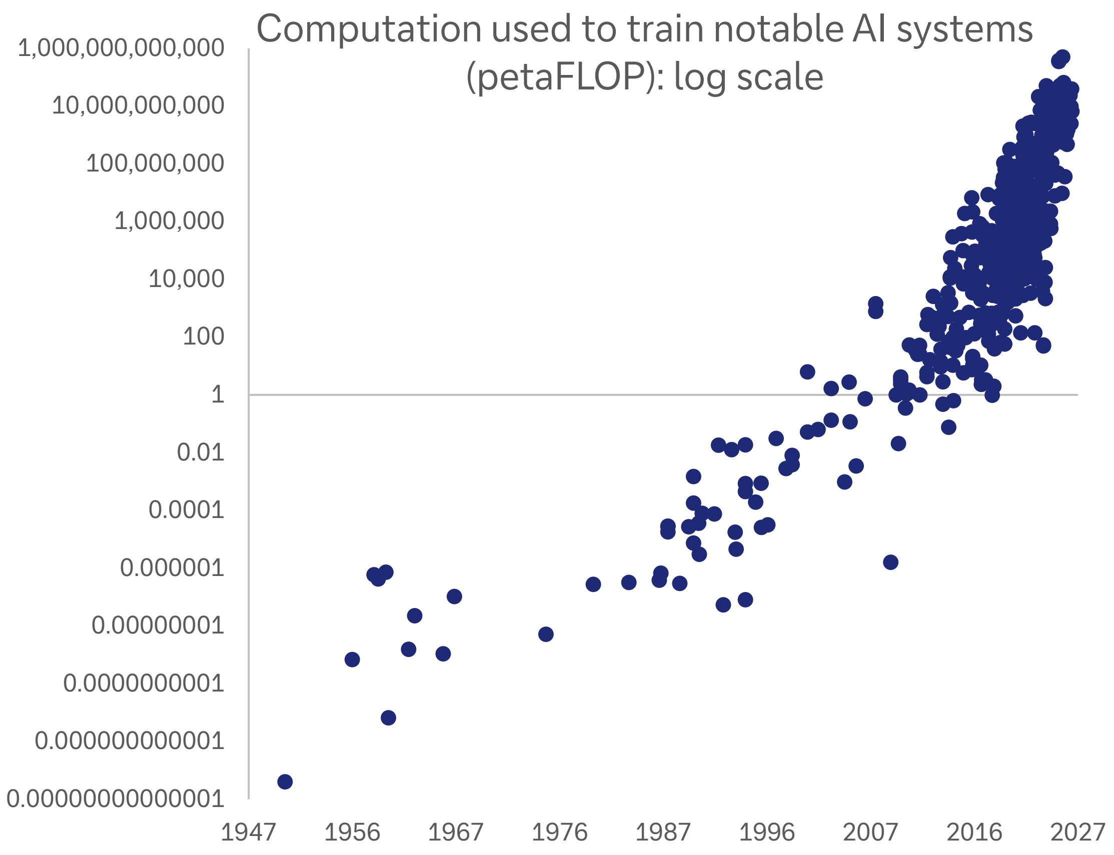
A Proposal for the

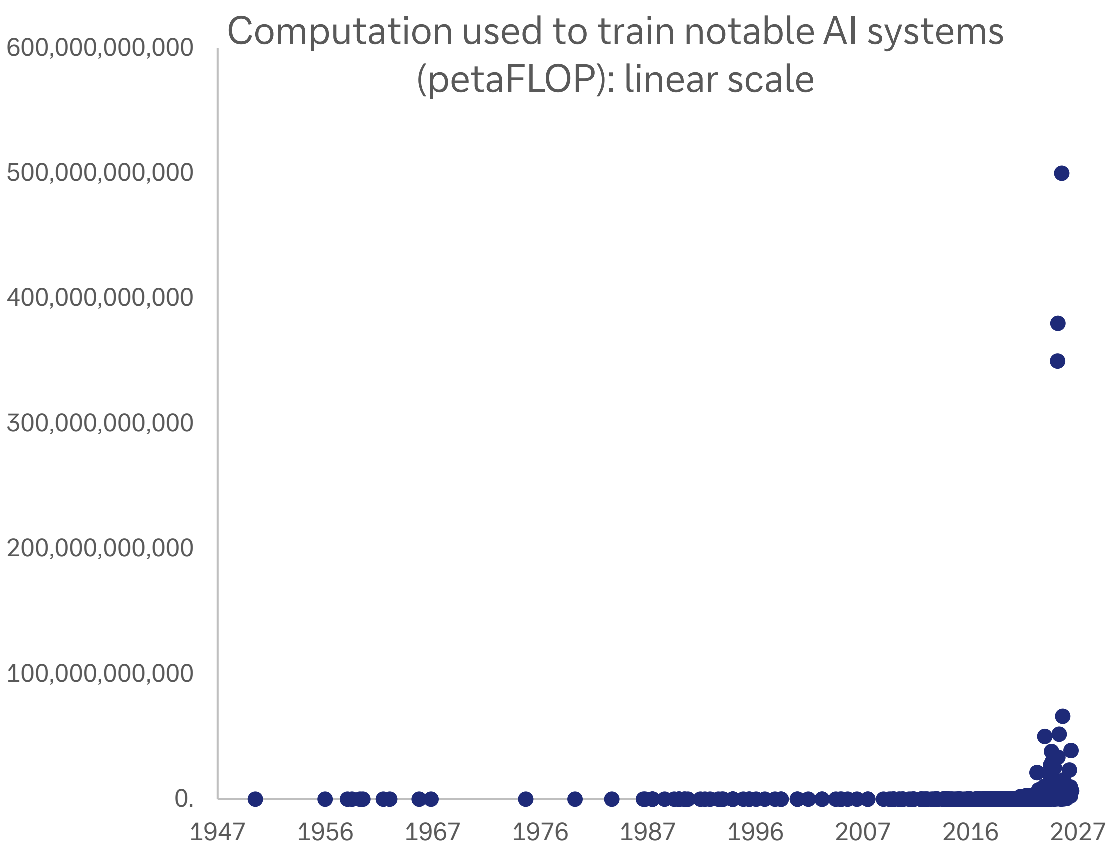
DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so pre-

1. AI progress is non-linear – and exponential growth is almost invariably underestimated

Compare what AI charts usually look like (on the left) vs what they actually represent (on the right)

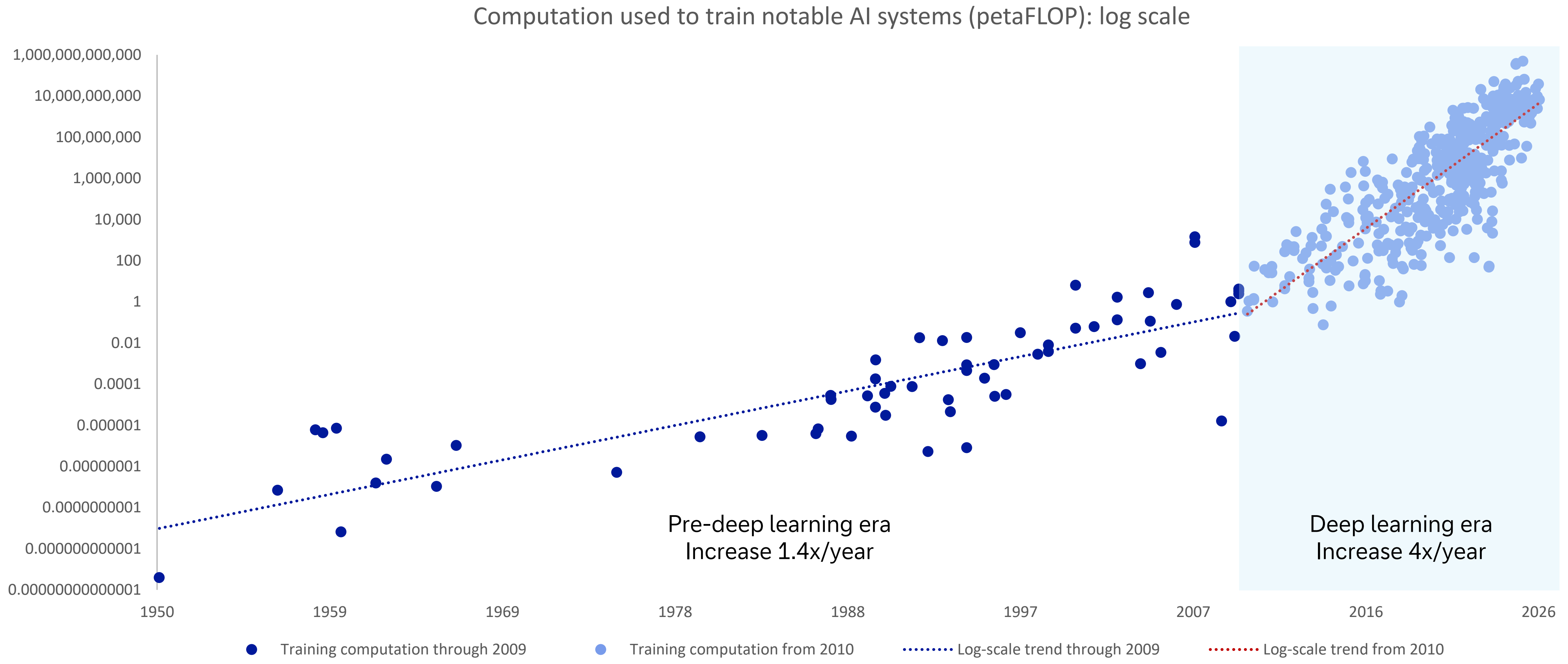




Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research. Note: One FLOP (floating-point operation) represents a single arithmetic operation using floating point numbers; one petaFLOP is one quadrillion FLOPs

2. The pace of progress exceeds Moore's law when many factors are improving at once

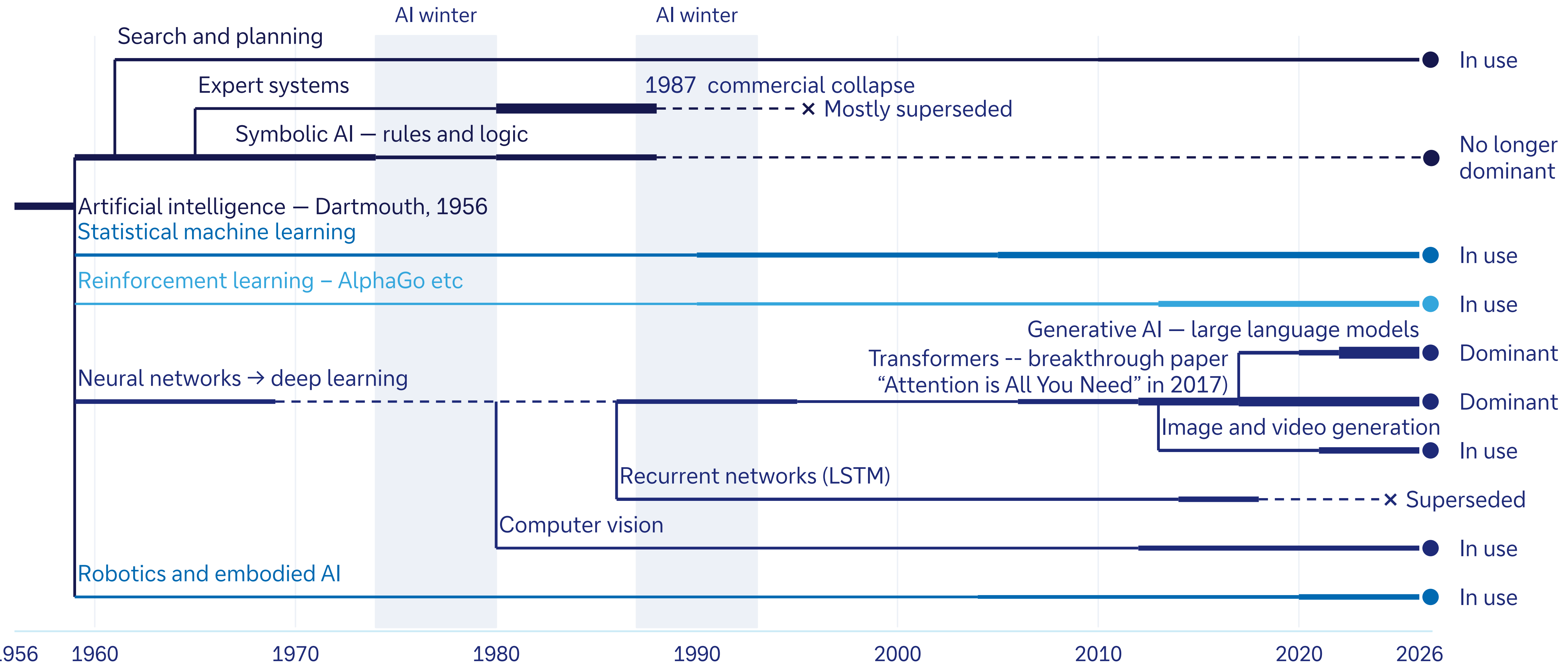
Conventional computing capability has doubled every 18-24 months but bigger and better systems, memory and algorithms have accelerated capabilities far beyond that



Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research

3. Today's leading tech may not be tomorrow's

AI has an ever-evolving family tree; large language models may give way to, eg, world models



Colour = lineage · Thicker line = more activity · Shaded = AI winters · × = superseded · ● = still in use

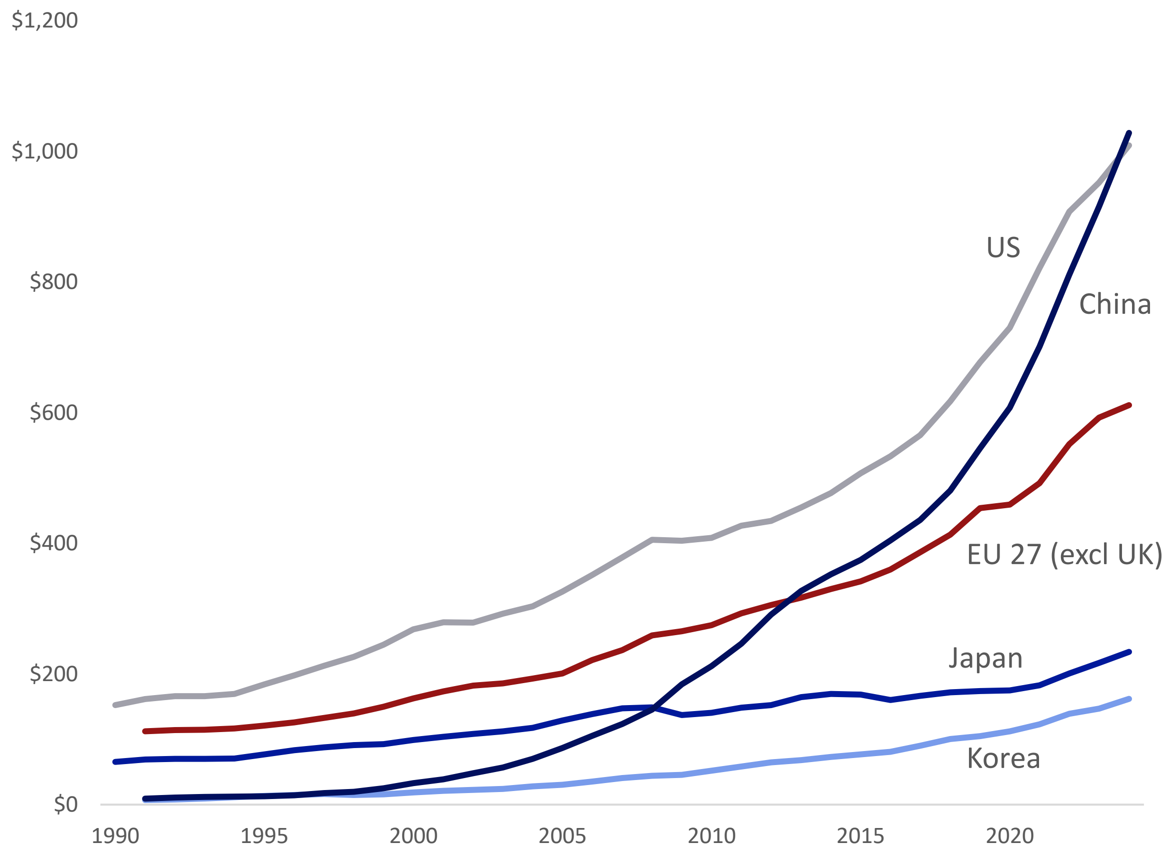
Dates mark impact, not first invention.

Source: Deutsche Bank Research

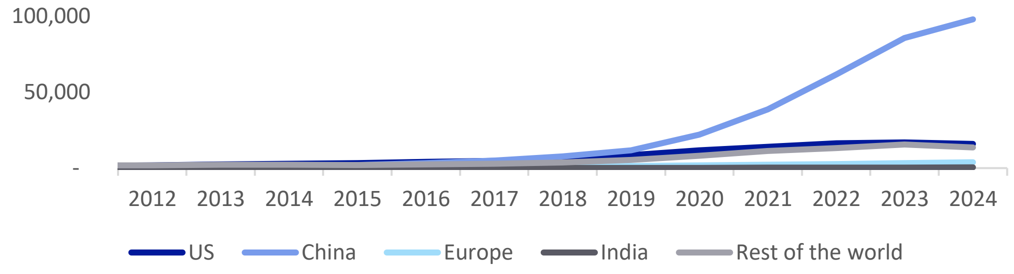
4. The pipeline matters: China's overnight success with DeepSeek was years in the making

Success of Chinese models reflects a multi-year push in R&D, overtaking the US in 2024

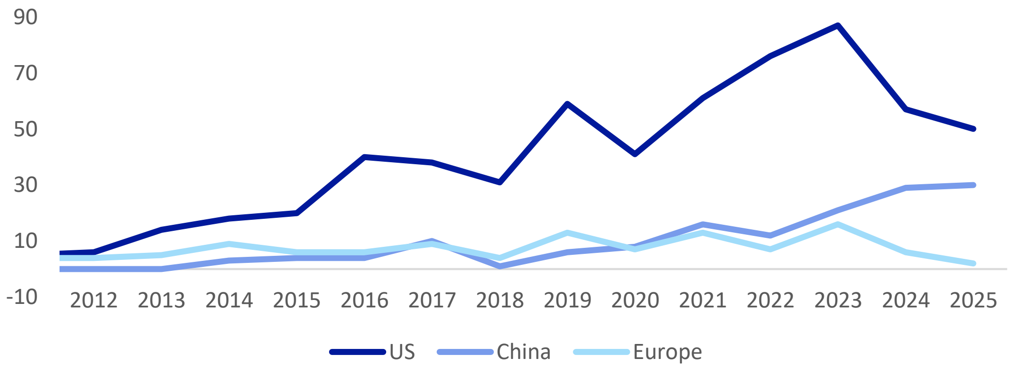
Gross domestic research and development spend by country, PPP converted, current prices (USDbn)



Number of AI patents granted by select geographic areas, 2012-2024



Number of notable AI models by geographic area (2012-2025)

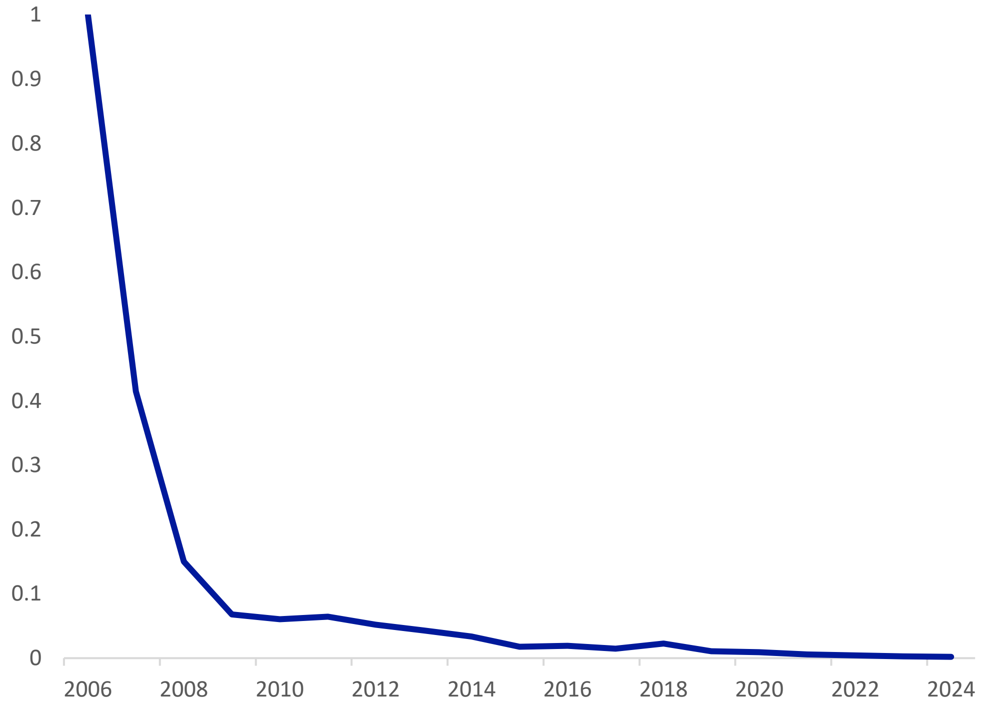


Source: OECD Data Explorer, Deutsche Bank Research

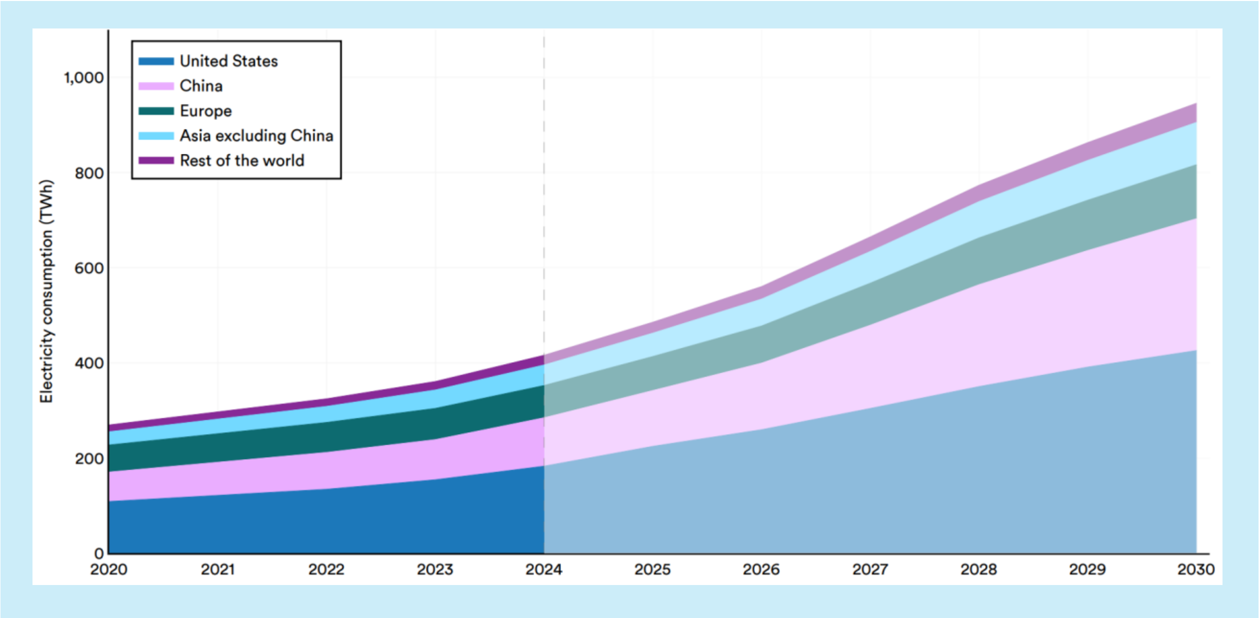
5. Making AI dramatically cheaper will likely increase, not decrease, spending

Jevons Paradox is alive: GPU computation cost has fallen by more than 99% since 2006, yet electricity consumption by data centres is set to double from 2024 to 2030 as demand surges

GPU chip computation cost index (2006=1)



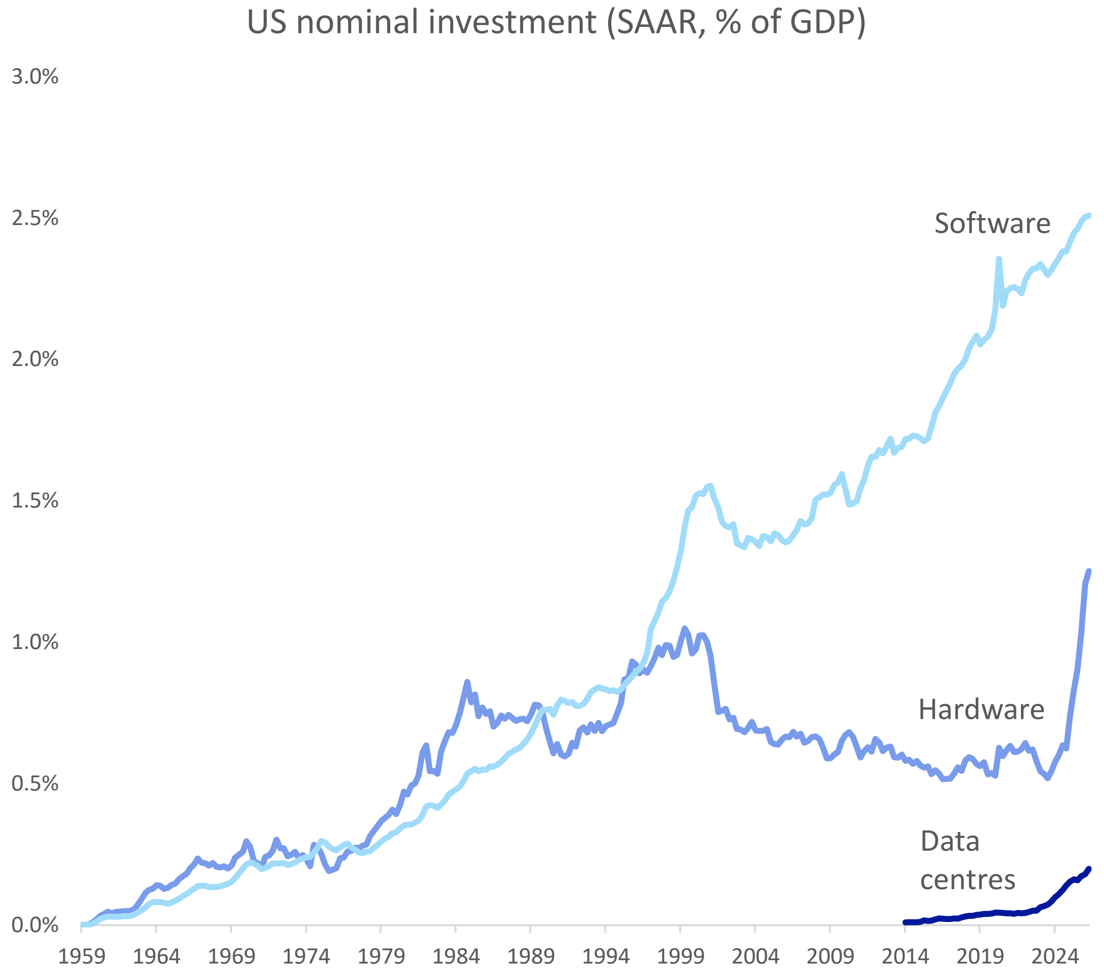
Data centre electricity consumption by region 2020-2030 (IEA projections)

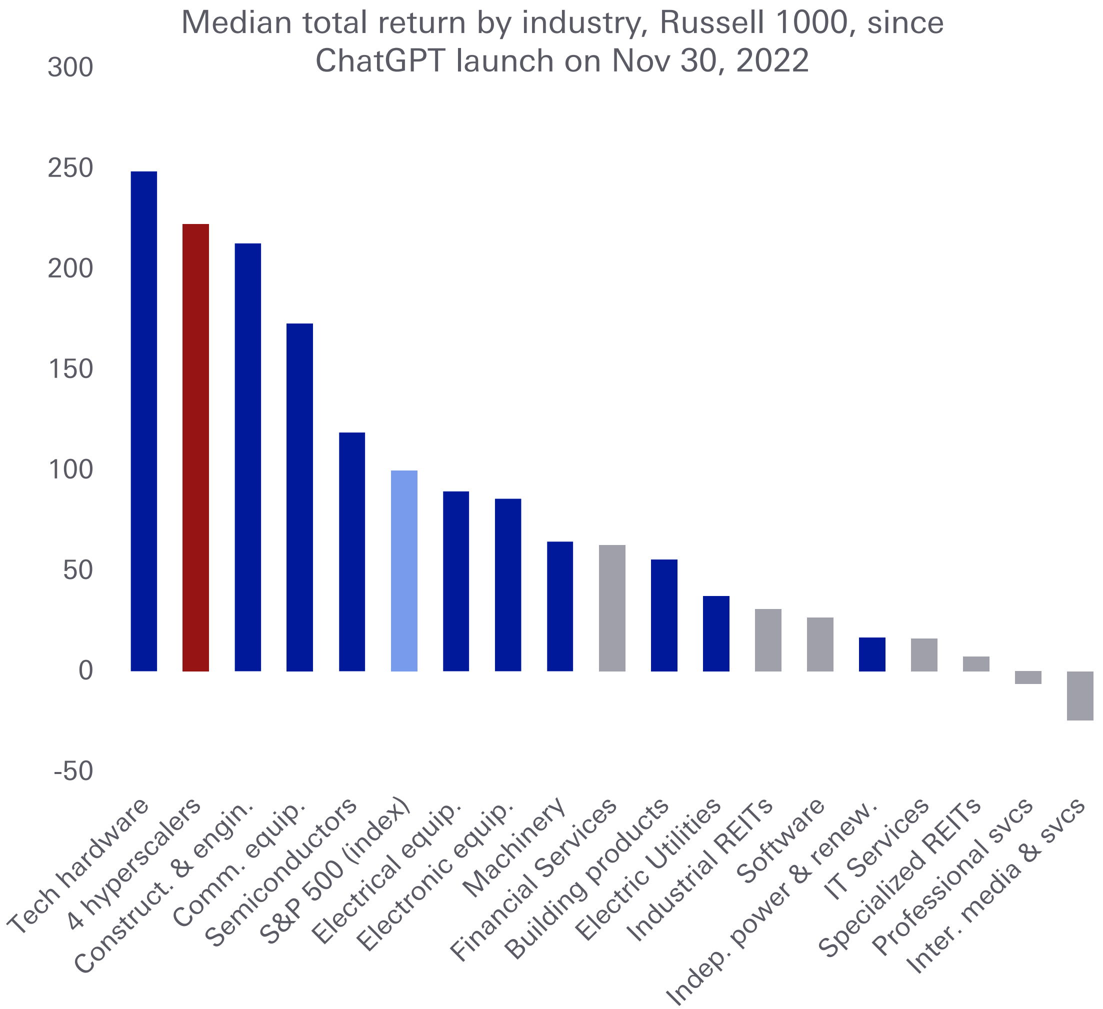


Source: International Energy Agency (2025), Stanford HAI AI Index 2026, Deutsche Bank Research

6. Software revolutions depend on heavy hardware investment

Hardware, construction, chip and equipment makers have led gains in the AI boom so far

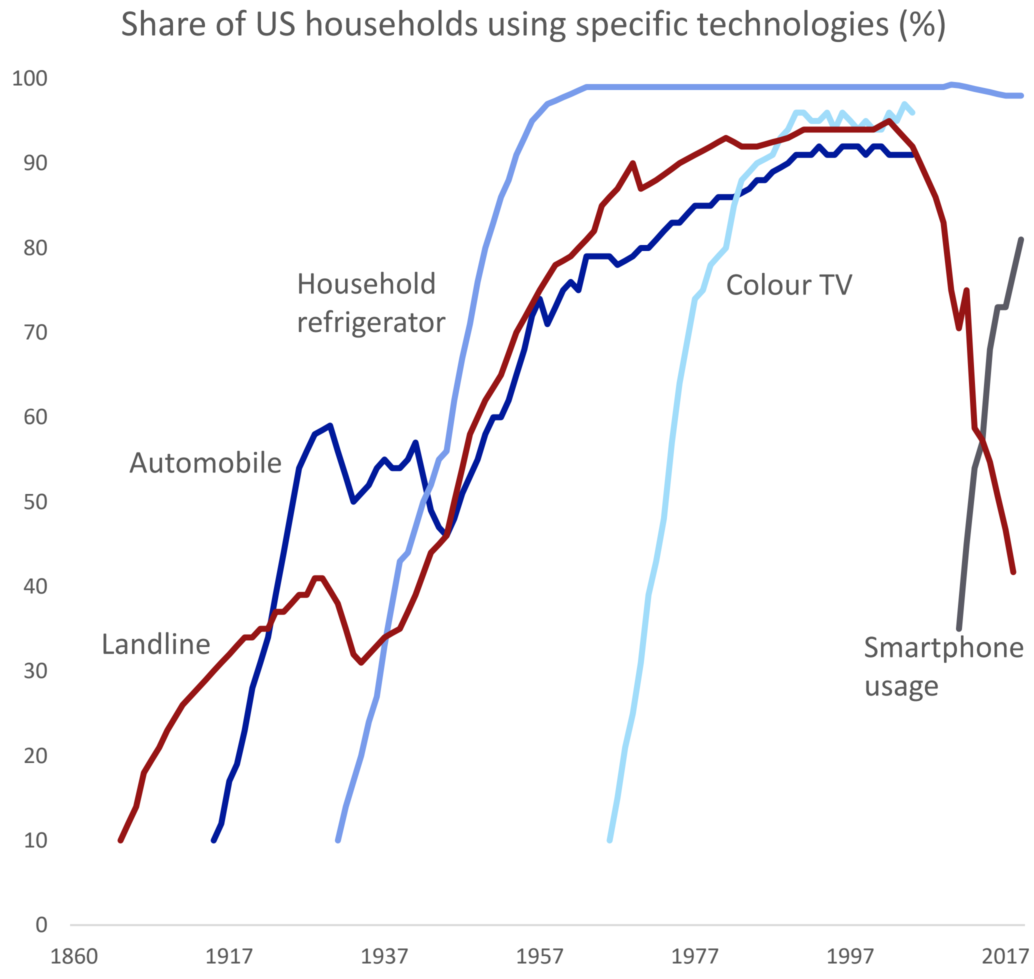


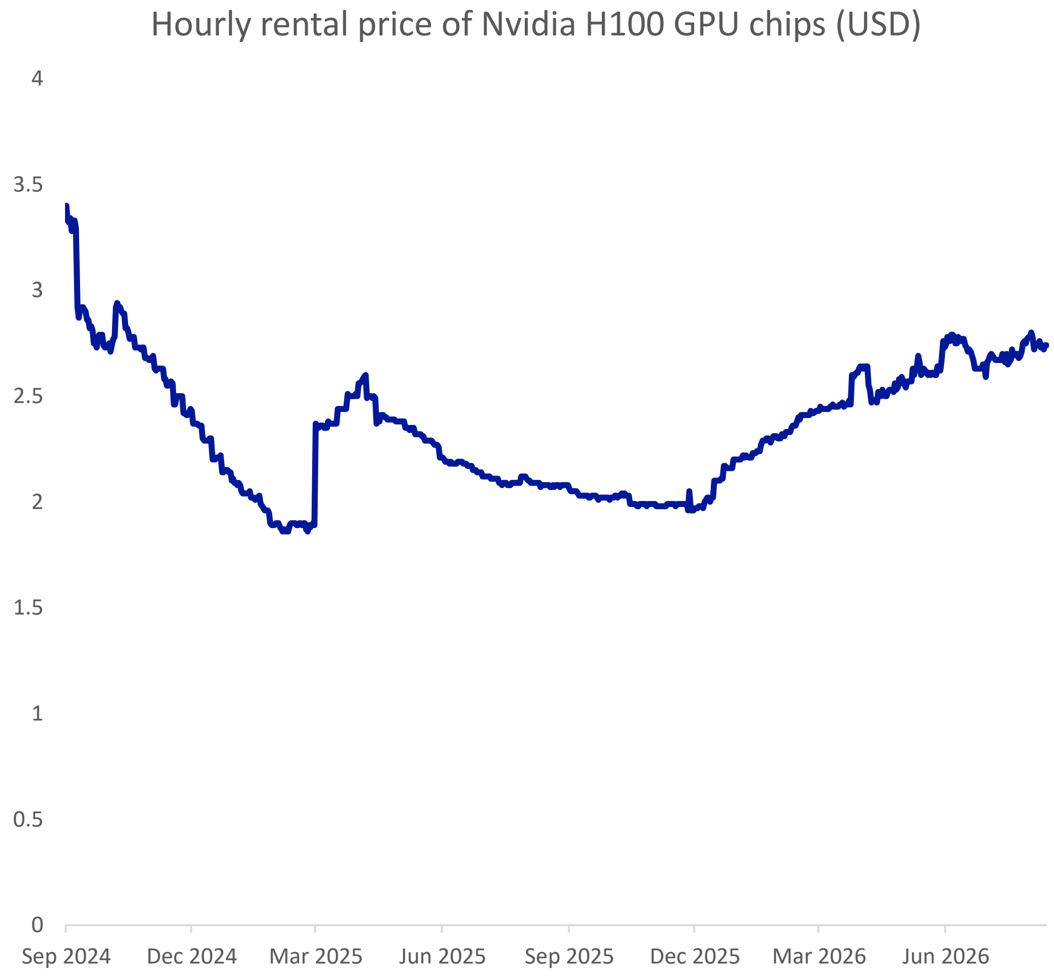


Source: Federal Reserve Bank of St Louis, Haver Analytics, Bloomberg Finance LP, Deutsche Bank Research

7. Hardware is often a bottleneck – but for once the delay is with producers not consumers

Past booms often relied on getting devices to consumers; now it's about getting scarce hardware online to serve them remotely

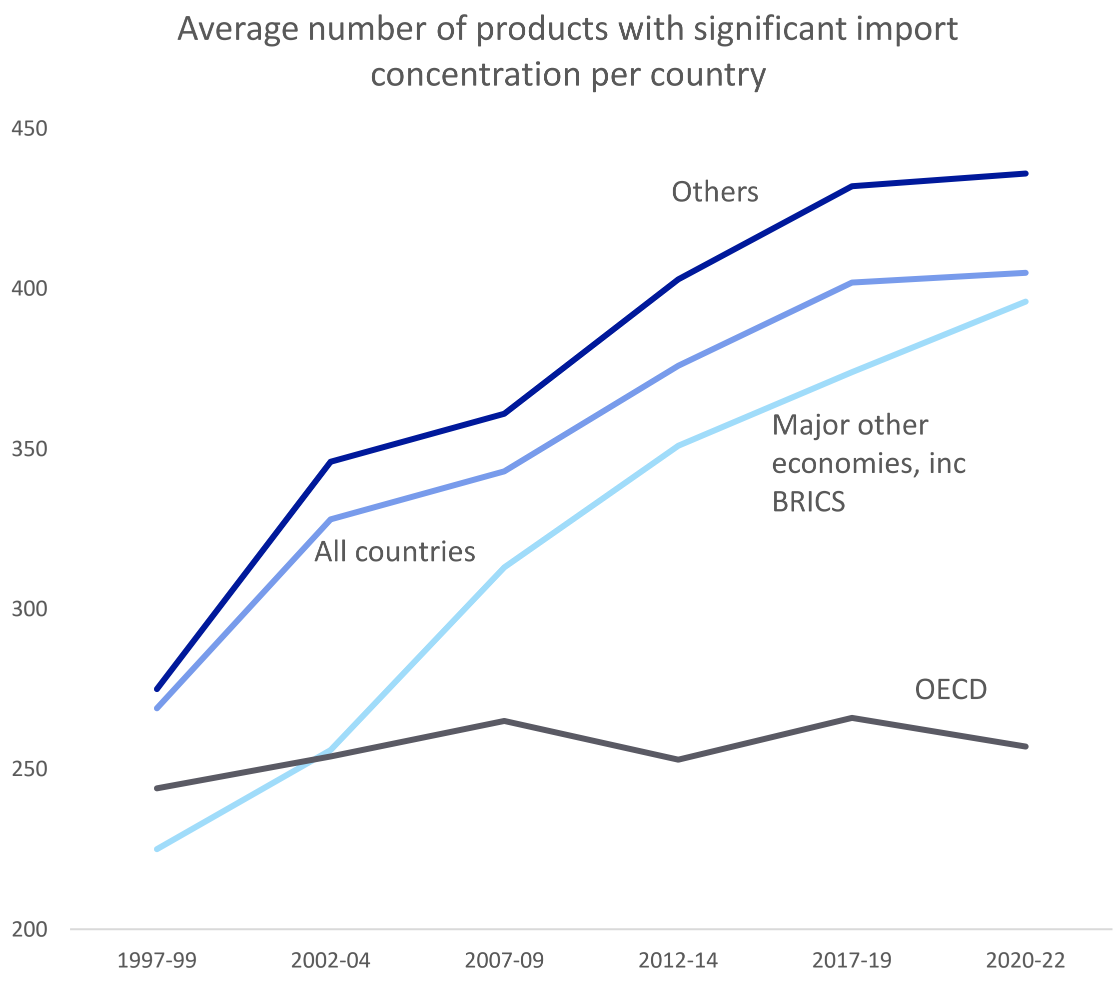


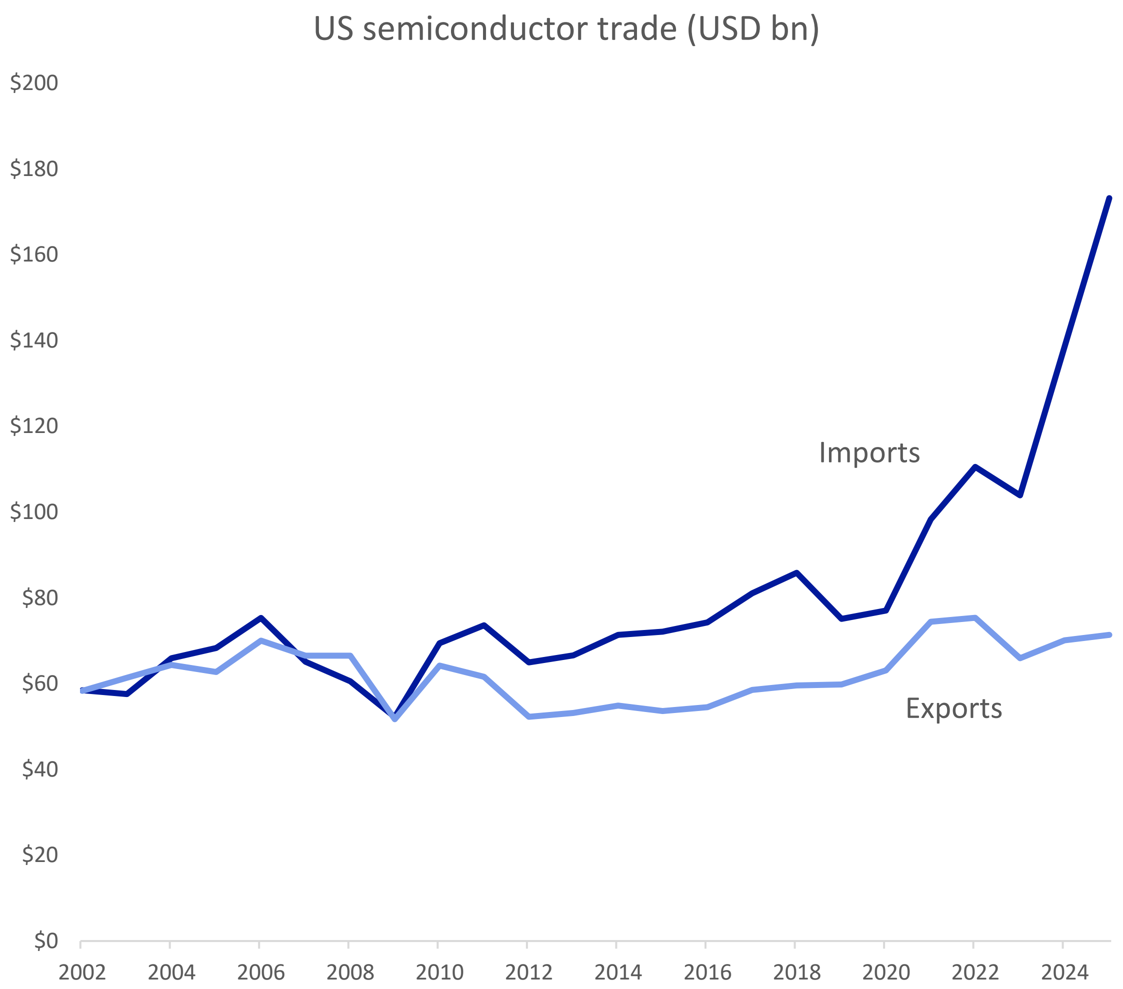


Source: Our World in Data, Federal Reserve Bank of St Louis, Haver Analytics, Bloomberg Finance LP, Deutsche Bank Research

8. Globalisation made tech cheaper – but also increased supply chain risks

There has been a surge in US imports of semiconductors as the AI boom has taken off

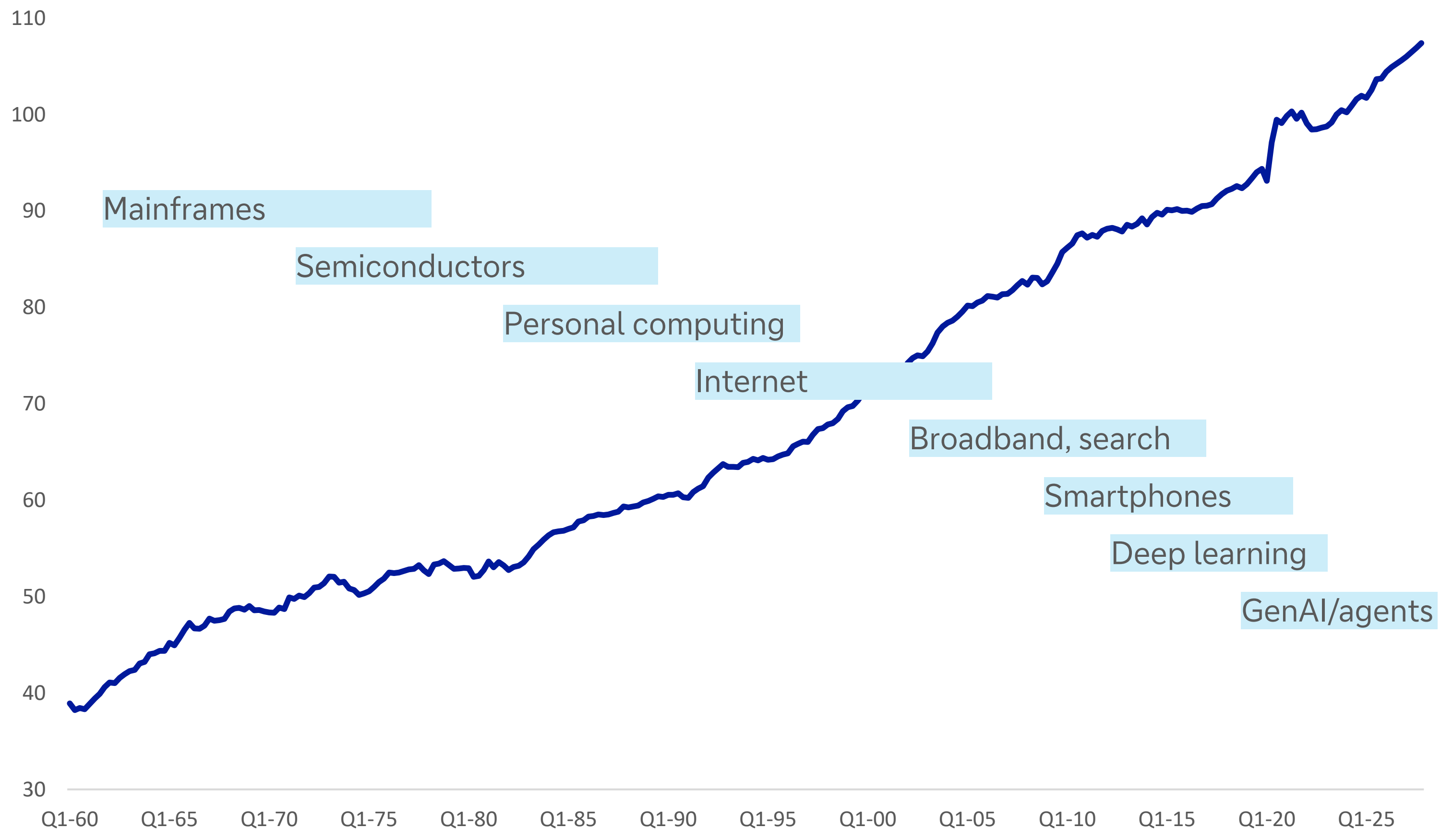




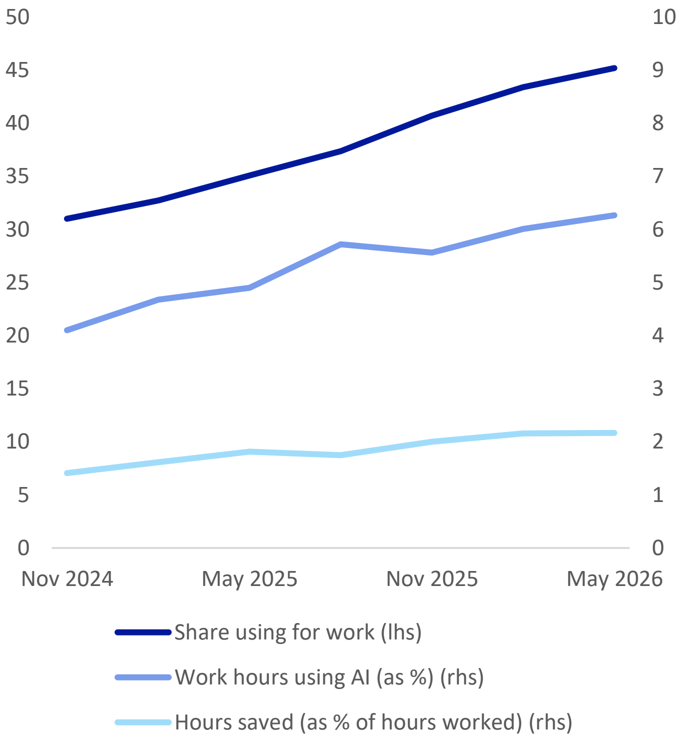
9. Tech revolutions take time to pay off, making them hard to see in long-term data

General purpose technologies have a productivity J-curve, generating costs before results

US productivity index since 1960 (2021 = 100)



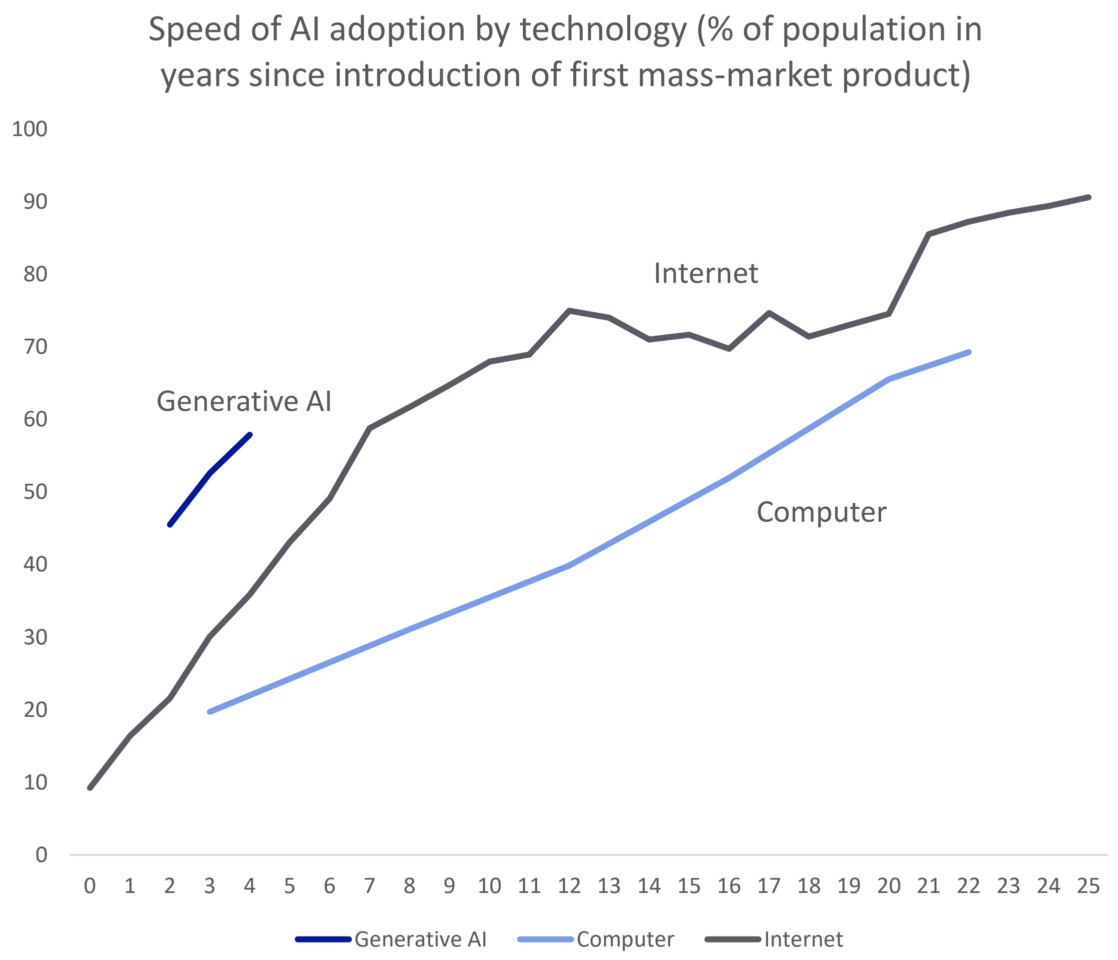
Fewer than half of US workers say they are using AI at work, with a mean of just 6% of time and saving 2% of hours

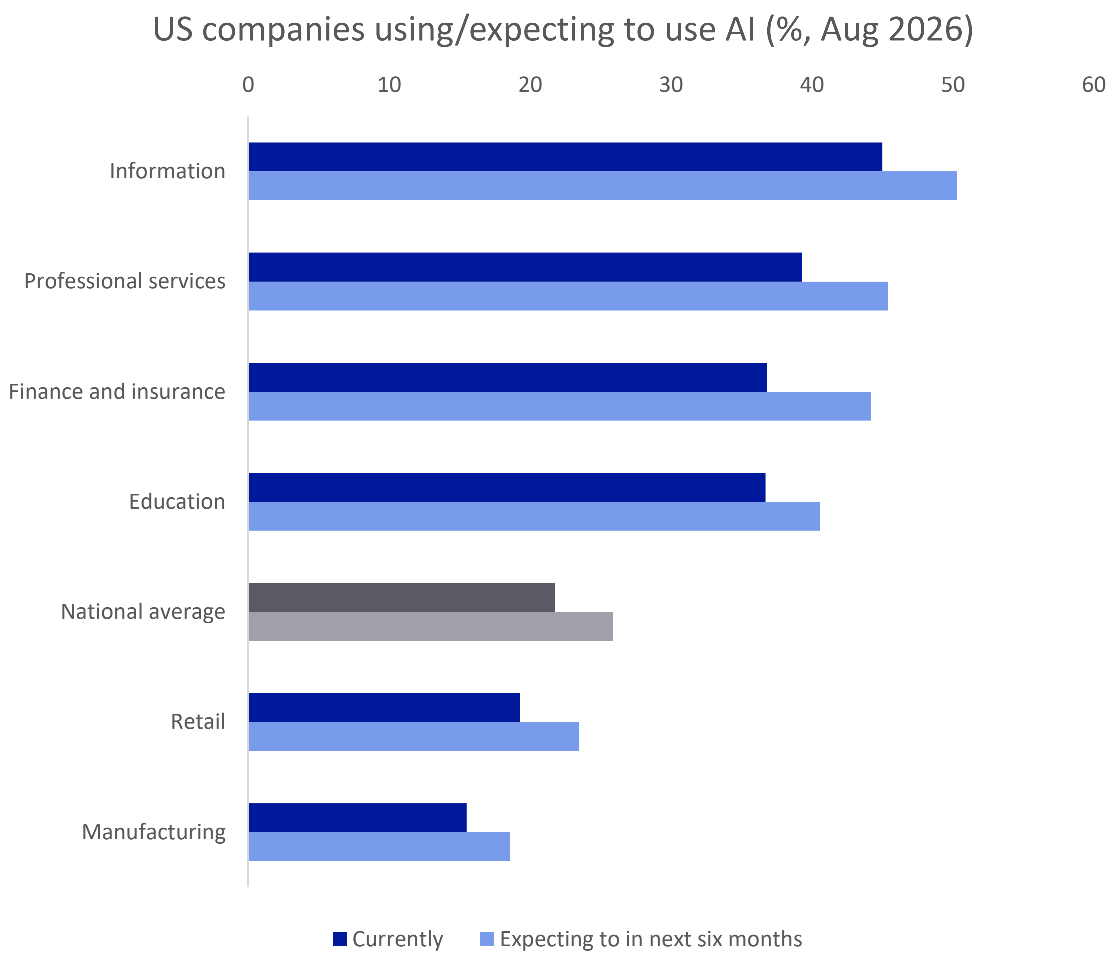


Source: OECD, GenAI Adoption Tracker, Deutsche Bank Research; blue bars in productivity chart give rough indication of notable tech eras

10. Consumers have adopted AI at a record pace but monetisable business use is slower

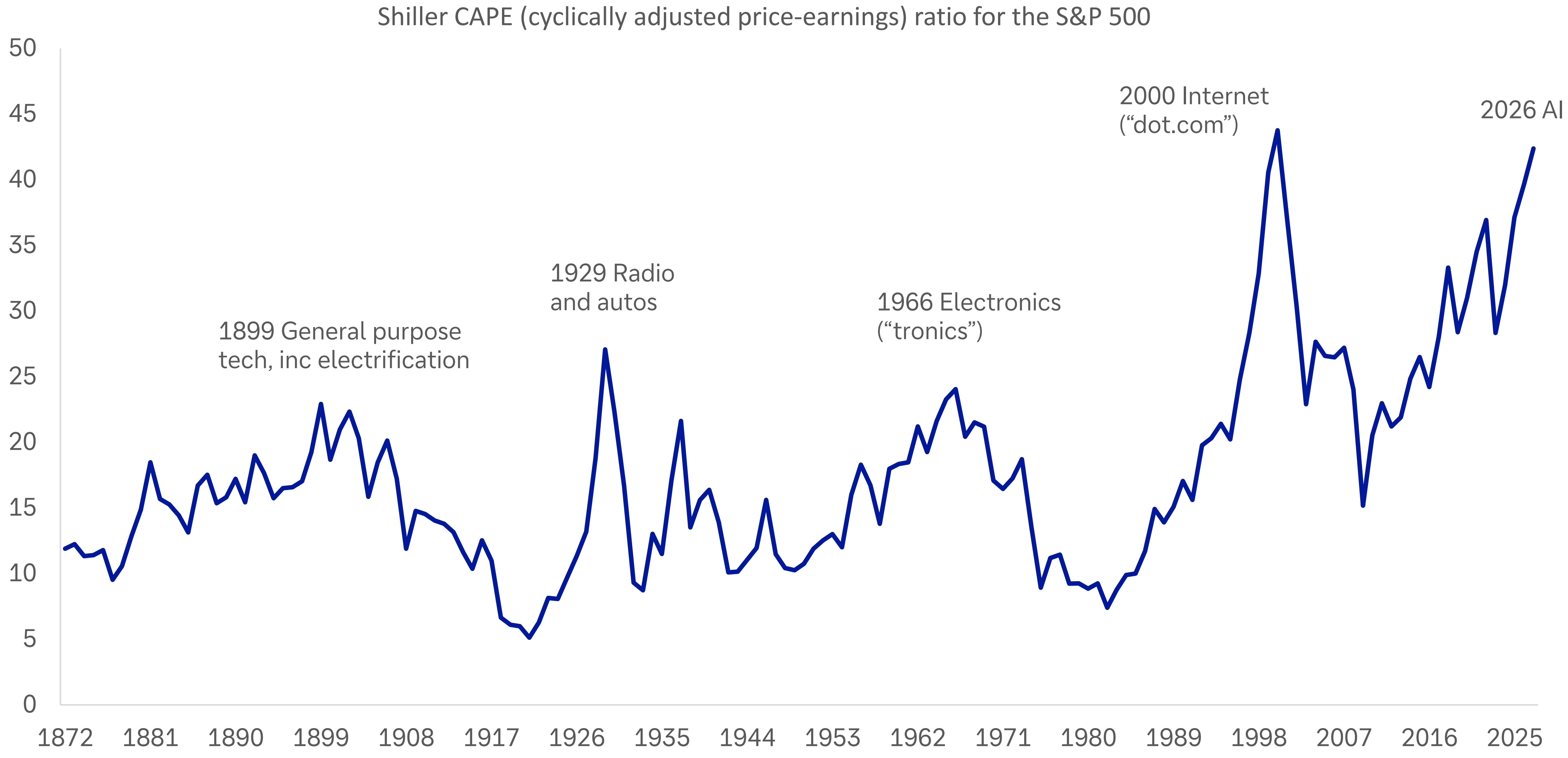
Downloading an app takes minutes; reorganising enterprises round new tech takes years





Source: GenAI Adoption Tracker, US Census Bureau, Deutsche Bank Research

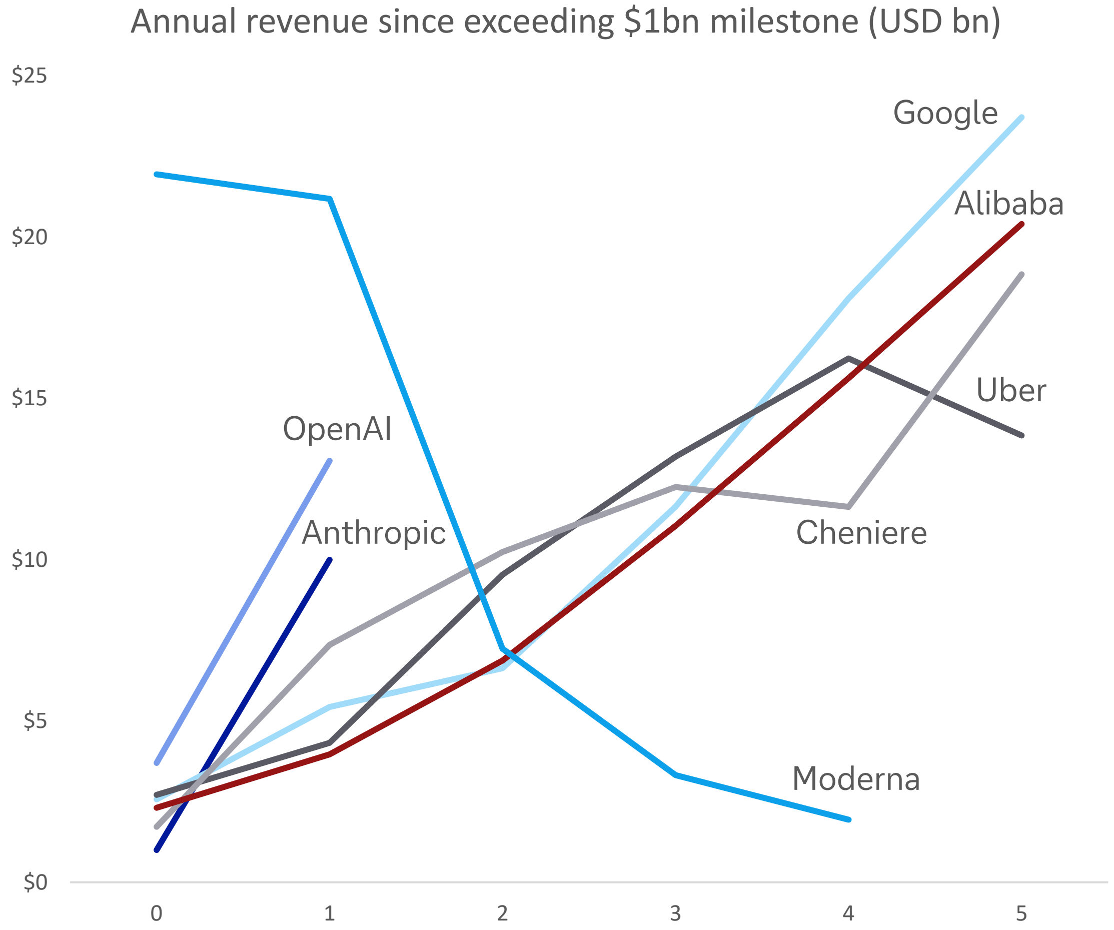
11. The biggest valuation booms are often tech booms – like today
Peaks have mostly coincided with transformative technologies



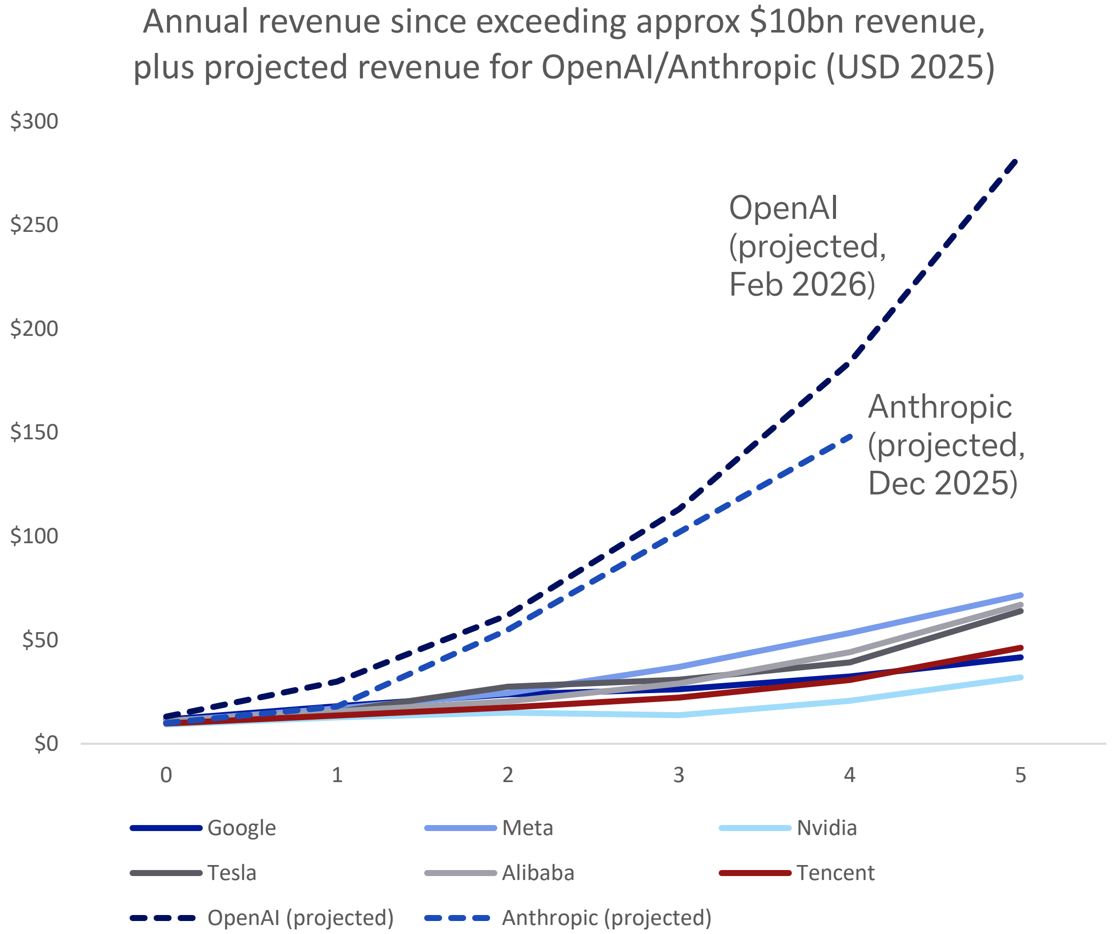
Source: Robert Shiller, Deutsche Bank Research

12. This time is different – but it will need to be even more different to meet projections

OpenAI and Anthropic are exceeding record revenue growth set by peers, at least for now



Source: Bloomberg Finance LP, media reports, Deutsche Bank Research. Note: Year 0 shown as first year revenue exceeded $1bn (2025 USD). For OpenAI and Anthropic that was reportedly 2025. Moderna nominal revenue spiked from $800m in 2020 to $18.5bn in 2021, due to Covid vaccine. Cheniere revenue spiked with 2022 energy crisis.

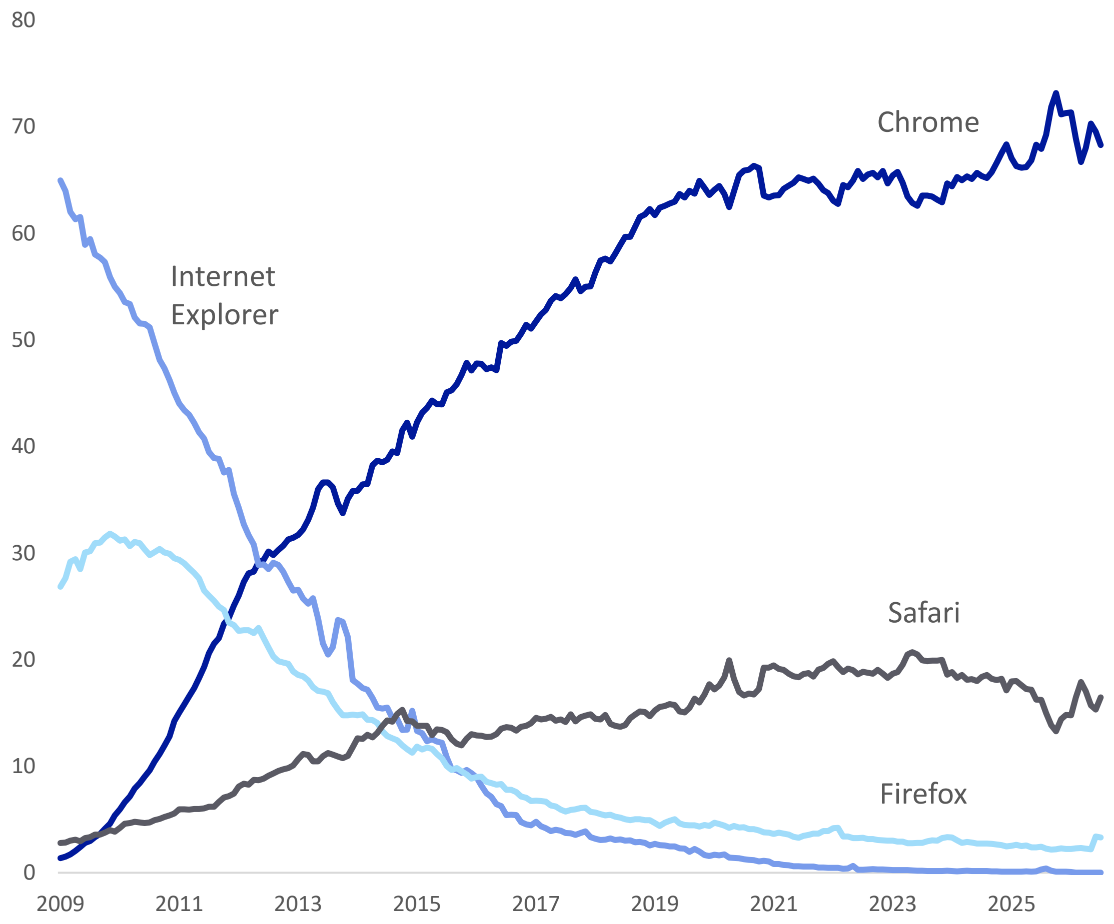


Source: Epoch, Bloomberg Finance LP, The Information (Feb 2026, citing recent updates to investors from OpenAI and Anthropic), Deutsche Bank Research. Note: Year 0 is first year revenue was approx $10bn (2025 USD)

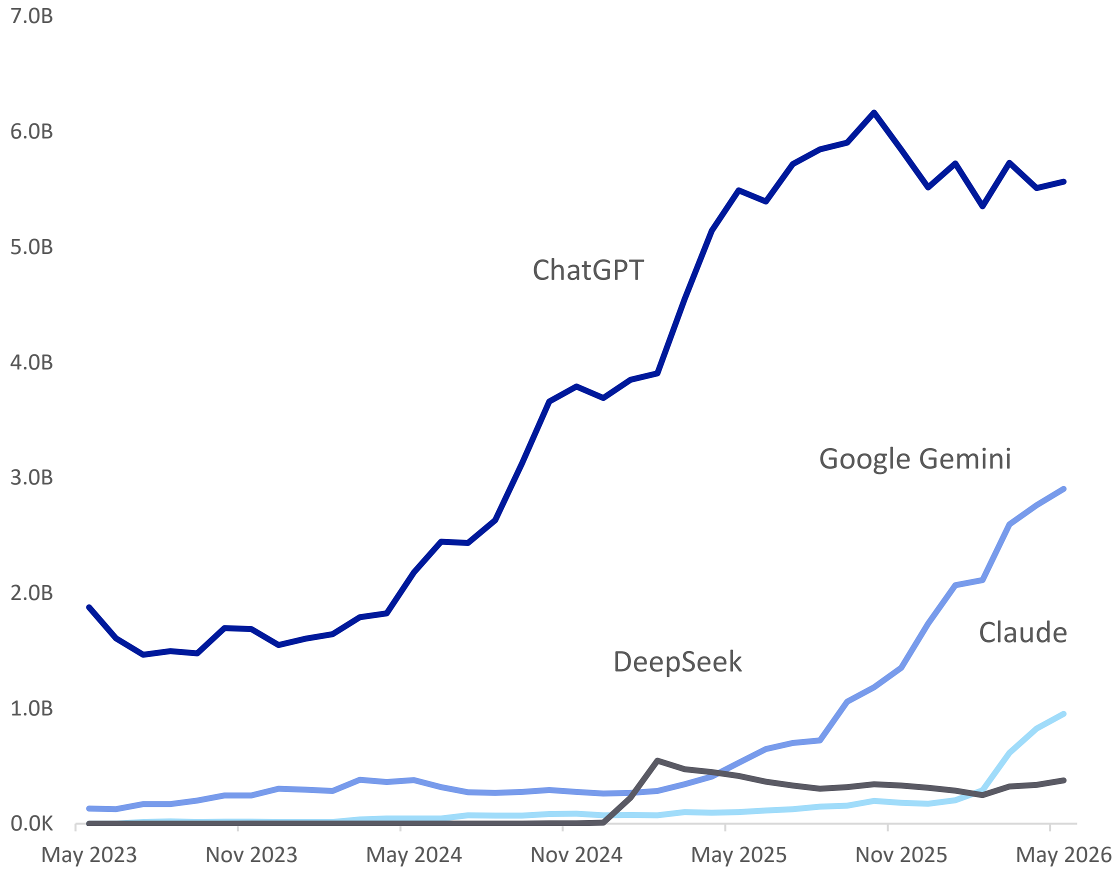
13. Early leads do not guarantee long-term winners

For example, top browser Internet Explorer overtook Netscape, then it too was overtaken

Browser market shares (%) (2009 to today)



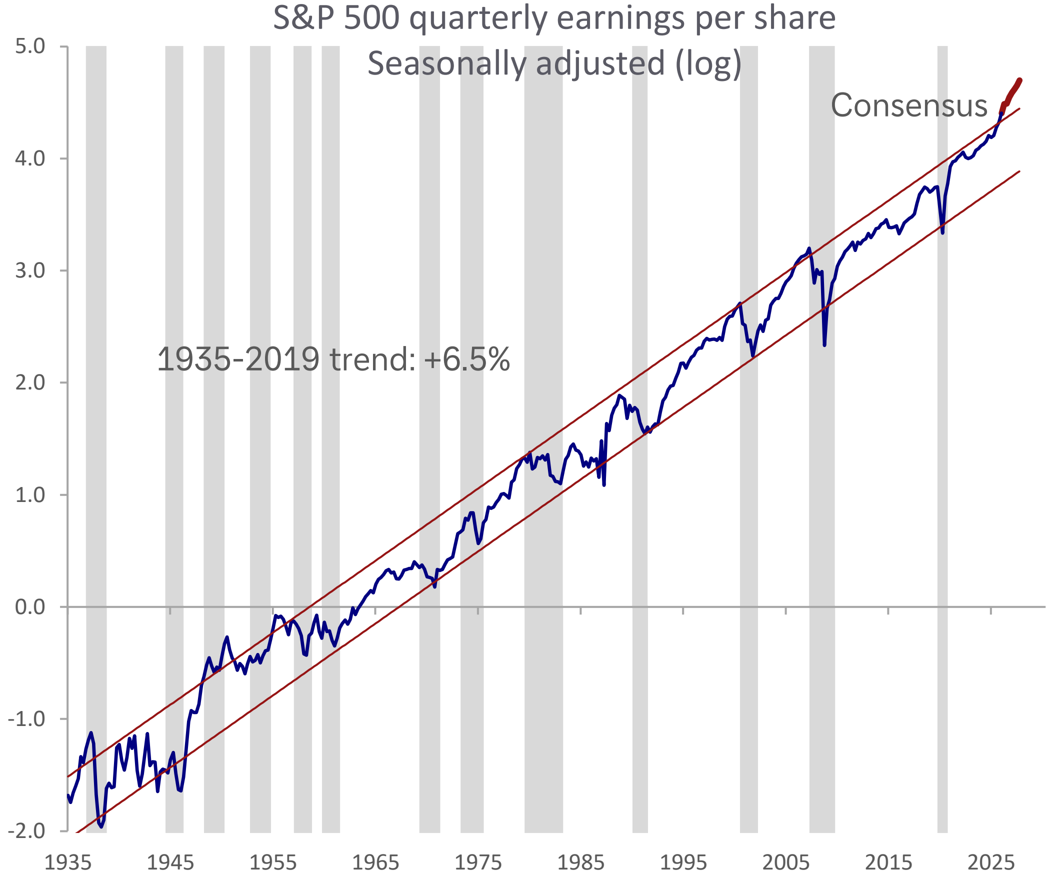
Top four generative AI platforms: monthly visits worldwide (desktop and mobile)

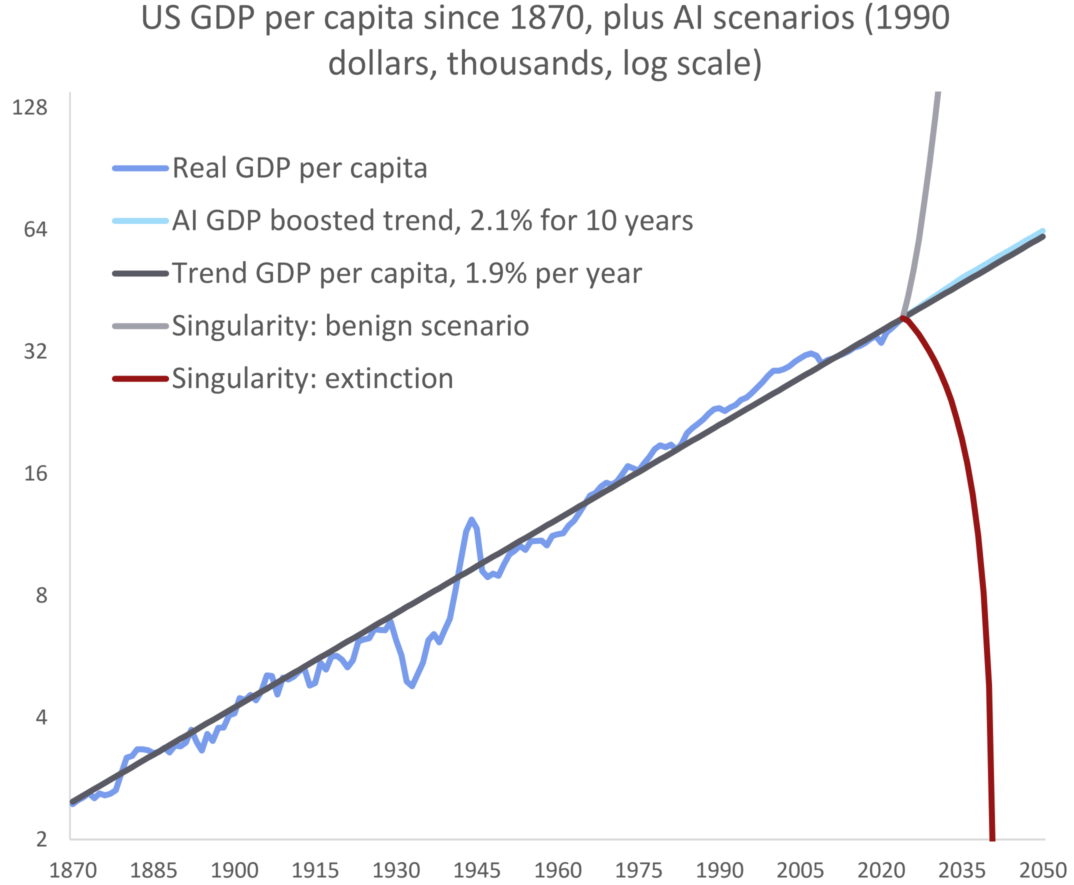


Source: Statcounter, Similarweb, Deutsche Bank Research

14. Long-term trends tend to persist through wars, recessions – and tech revolutions

Even so, some are considering more extreme scenarios this time… from utopia to extinction





Source: Bloomberg Finance LP, Federal Reserve Bank of Dallas, Deutsche Bank Research. Grey shaded bars in S&P quarterly earnings chart represent recessions

Appendix 1

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Deal or no deal, investors are both concerned and excited

August 11, 2026

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Executive summary

Five months into the US-Iran conflict and the strength of corporate earnings makes it seem like the optimists have won. But things are more complicated. This report analyses proprietary data from our dbDataInsights team and shows that the issue of investing against the backdrop of a volatile energy market is highly polarising for people. Indeed, of all the investment themes we track, energy has one of the highest rates of both optimism and pessimism.

So, we have an odd situation: On one hand, global corporate earnings are incredibly strong despite the energy shock this year. On the other hand, the Iran conflict is still a massive problem for the world economy. An amicable agreement thus remains elusive.

Three findings from our analysis stand out.

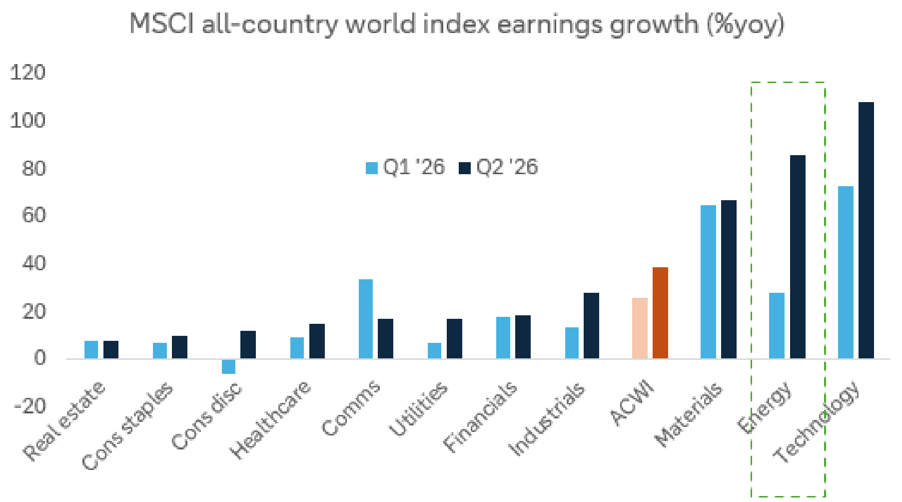
— Energy – whether measured through prices or transition – is now one of the most-cited investment themes for US investors behind AI, and is prompting portfolio changes on par with technology and geopolitics.
— Energy prices carry one of the highest perceived risks of any theme we tested among UK investors.
— Our megatrend model shows that energy disruption (as a medium-term trend) has remained a positive force for the world economy this year. That pattern closely echoes the 1970s oil shocks.

Energy shocks are unambiguously bad for consumers and growth in the short term. But over longer time frames, our model shows that energy disruption has tended to be a net positive for markets and economies because it accelerates diversification of supply and pushes economies to generate more GDP per unit of energy consumed. We see early evidence of a similar dynamic beginning today.

A serious issue ... but surprising resilience

The closure of the Strait of Hormuz has not been the economic catastrophe that many expected five months ago. In fact, corporates earnings have just had one of their best quarters outside of recession recoveries. For the companies in the MSCI all-country world index, median earnings growth is running at 16.5% yoy in Q2, well above the 14.7% in Q1. Energy has been a big part of that, with earnings growth, at a sector level, surging 86% in Q2, about triple the growth recorded in the prior quarter.

Figure 1: Earnings growth has been very strong in Q2



Source: Bloomberg Finance L.P., Deutsche Bank Asset Allocation

Economies have fared less well than companies, but they remain remarkably resilient given the harsh predictions that came from some quarters in early March. Of course, stockpile releases and other buffers have helped smooth the disruption – and as these wear down further, larger problems may arise.

So, now we have a curious juxtaposition of outlooks. On one hand, corporate earnings are proving resilient and the long-term focus on energy security has helped make many of the associated equities good investments. On the other hand, the Iran conflict is still a massive problem for the world economy.

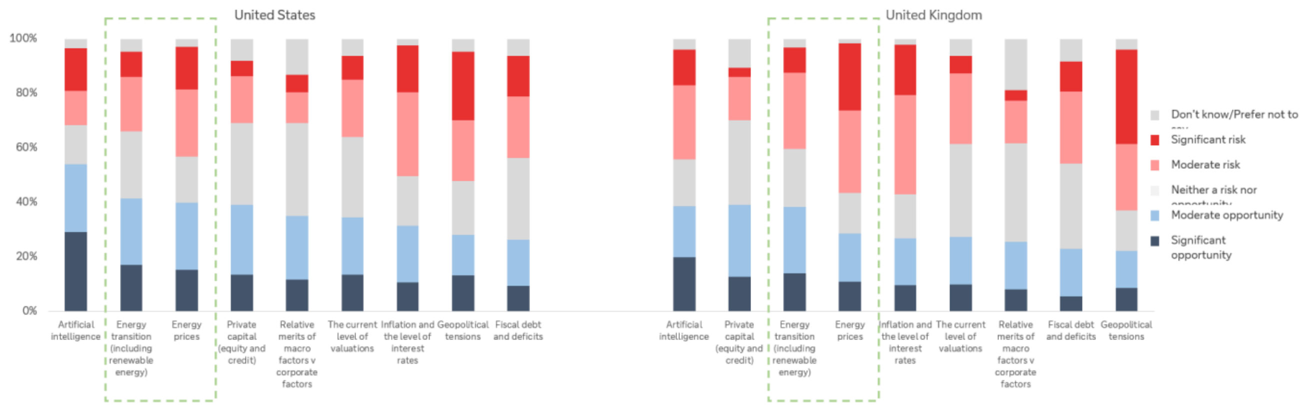
What investors think about the effect of energy on their portfolio

Our dbDataInsights survey on the topic was carried out in April and May of this year and polled a representative sample of people in the US and the UK. The responses indicate that energy has become one of the two dominant investment themes of 2026 – alongside, and closely intertwined with, artificial intelligence.

Energy is a high-conviction theme – in both directions

When we asked investors whether they see various investment themes as a risk or an opportunity for 2026, energy transition and energy prices ranked just behind artificial intelligence as the most-cited opportunities in both the US and the UK. The below chart shows a selection of the themes.

Figure 2: Survey question: "For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026"



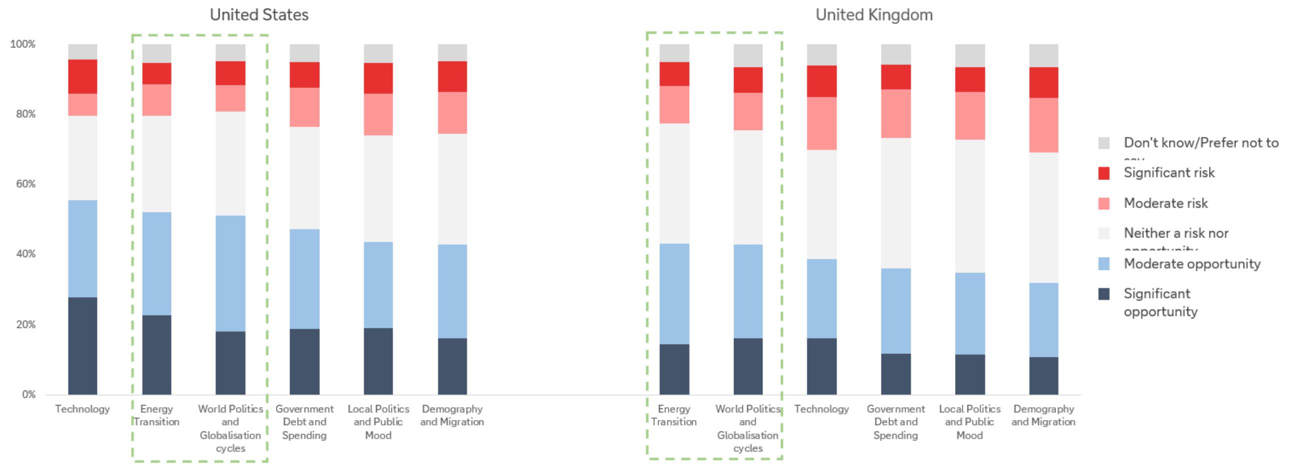
Source: dbDataInsights, Deutsche Bank

Energy is already changing portfolios

Conviction about energy appears to be translating into action – particularly when we think about it from a medium- or long-term point of view. To assess this, we fell back on our megatrend model that assesses the development of the six key megatrends that are driving the world's economies and markets.

When we asked investors how likely they are to change their portfolio over the next 12 months because of each megatrend, a dominant theme was energy and geopolitics as the below chart shows.

Figure 3: Survey question: "Within the next 12 months, how likely are you to make changes to your investment portfolio due to the following thematic megatrends?"

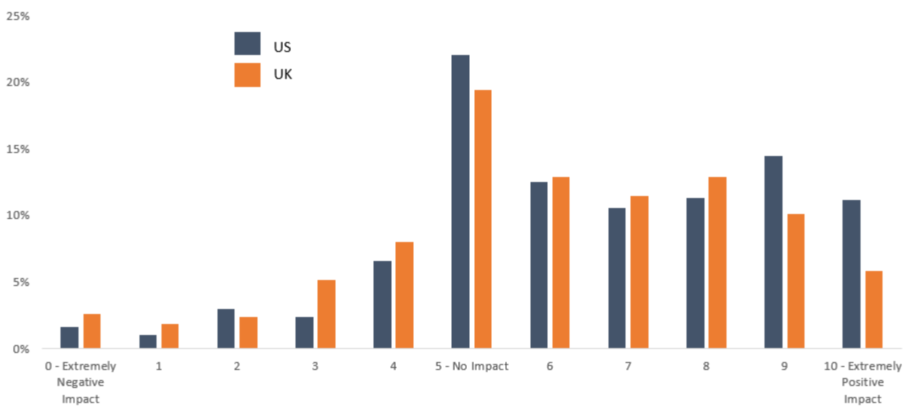


Source: dbDataInsights, Deutsche Bank

Taken together, the two survey questions tell a consistent story: energy has moved from a background, structural megatrend to one that investors are actively trading around, much as technology has been for the AI boom. The difference is that, unlike technology, energy's near-term path is being set as much by missile strikes and shipping insurance premia as by fundamentals.

Investors appear to sense this: when we asked how the energy transition will affect their main portfolio over the next three years, responses skewed firmly positive.

Figure 4: Investors' anticipated impact of the energy transition on their main investment portfolio over the next 3 years

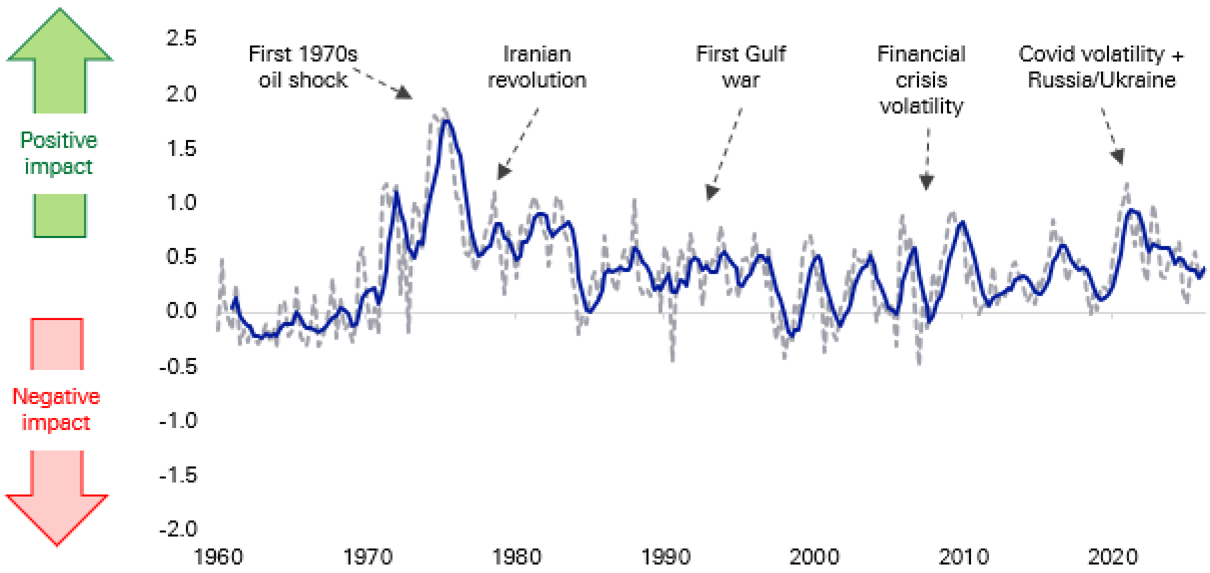


Source: dbDataInsights

Energy as a megatrend: why the medium-term signal is so positive

Energy is one of the six long-term trends that feeds into our megatrend model that uses almost 100 data series since 1957 to track their impact on economies and markets. Energy is a nuanced megatrend as, unlike several others, there is a big difference between the effect of short-term disruption and long-term gain. Indeed, prior periods of energy disruption have been incredibly good for the long-term energy trend, even if the short-term disruption was incredibly painful for consumers and economies.

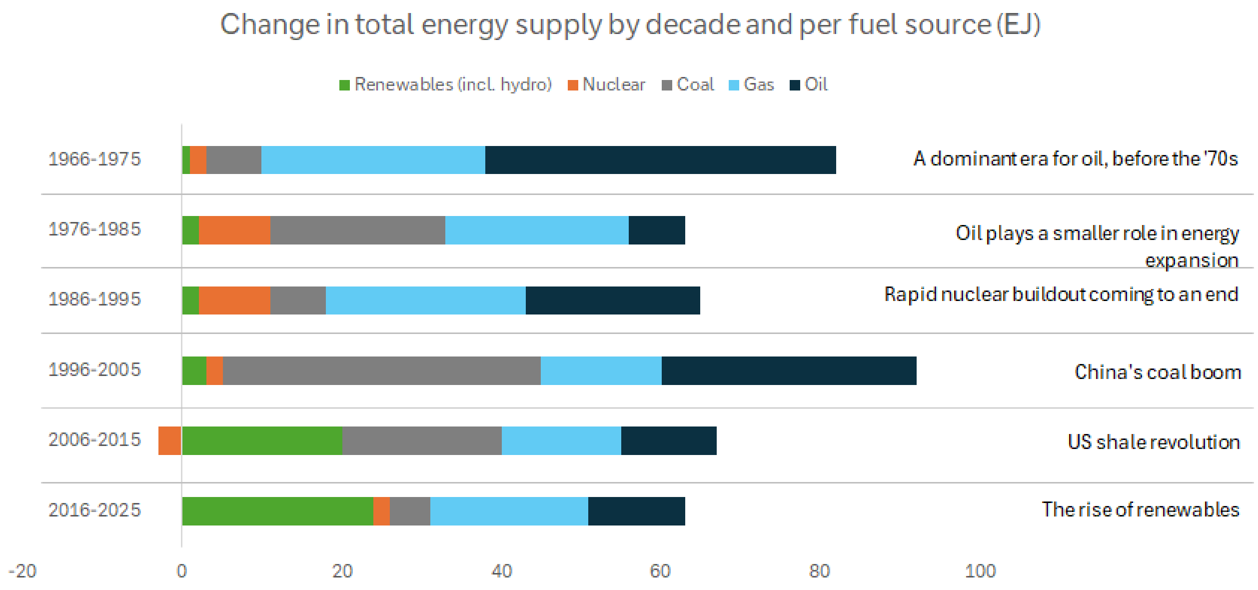
Figure 5: Our energy megatrend indicator has generally been positive as the world's economies diversify their supplies at the same time that innovation makes the economy more energy efficient



Source: Factset, Bloomberg Finance L.P., Fineon, EIA, Deutsche Bank Research

Viewed through a medium-term lens, the last five months of conflict in the Middle East can be viewed as likely being a long-term positive for economies and markets. That is because the Iran war is accelerating exactly the kind of diversification our model rewards: European and Asian buyers have moved further into US, Guyanese and West African crude and LNG since the Strait's closure; several G20 governments have fast-tracked strategic reserve releases and non-Gulf supply agreements; and renewed urgency has attached to the 'friend-sufficient' push for critical minerals and energy inputs that began during the US-China strategic rivalry.

Figure 6: Shocks over the decades have defined several eras for the evolution of the energy mix



Source: Energy Institute

The 1970s is something of a parallel

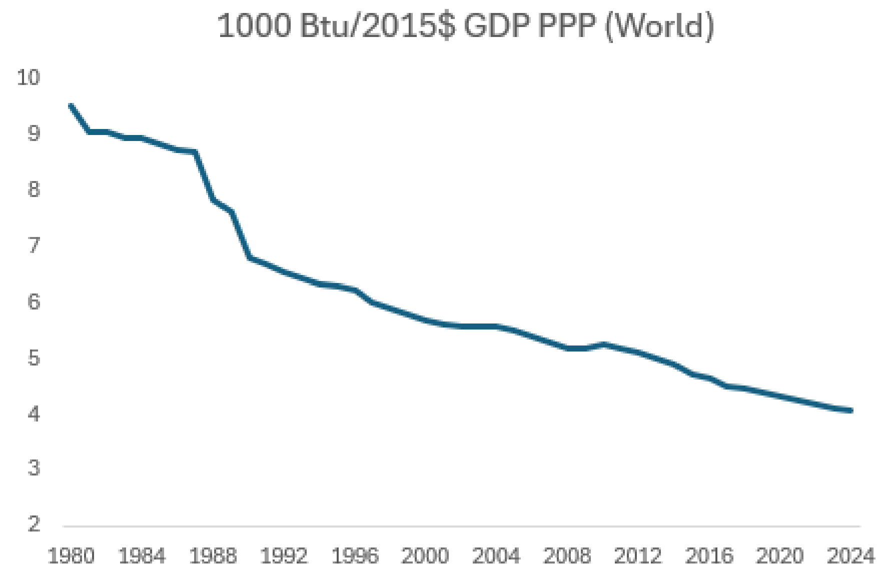
The closure of the Strait of Hormuz in 2026 and the disruption to Russian gas flows into Europe in 2022 both echo the twin oil shocks of 1973 and 1979, when OPEC's move to set its own prices followed by the Iranian revolution, sent crude to ten times its early-1970s level and permanently altered the trajectory of global oil demand.

Global oil consumption growth slowed from 7.8% before 1973 to 2.2% pa for most of the rest of the decade (on calculations from the Energy Institute). It outright declined after the second and took a decade for global oil consumption to recover to its 1979 level.

Whilst the 1970s shocks created havoc for both individuals and economies, they accelerated the pace at which the world changed how it generated economic output. The energy intensity of GDP fell 3.3% in 1980 (the largest single-year improvement in the Energy Institute's data series). That adjustment reflected both behavioural change and, over the following decade, structural investment in fuel efficiency and the phase-out of oil-fired power generation.

Since 1980, the energy intensity of GDP has roughly halved as shown in the following chart.

Figure 7: The energy intensity of GDP has been steadily falling – partly because of efficiency and innovation



Source: EIA

This trend is captured in our megatrend model which shows the medium-term payoff from decades of energy disruption. Those forces continue today. Every episode of chokepoint risk, from Ukraine in 2022 to Hormuz in 2026, adds another increment to the diversification of global energy supply and another incentive for energy innovation across economies.

Conclusion

Five months into the US-Iran conflict war, investors are highly polarised about the investment merits of investing against the backdrop of energy volatility. However, those that are positive are very positive. Our megatrend model suggests that reaction is rational: like the 1970s oil shocks, the medium-term legacy of this conflict is likely to be faster energy diversification and a further, durable improvement in the energy intensity of GDP, even though the near-term path for prices is likely to stay volatile.

As a result, energy security and supply diversification should keep attracting capital regardless of where the oil price settles. The beneficiaries are likely to cascade up and down the value chain including, LNG infrastructure, non-Gulf production, renewables, strategic reserves and grid resilience as countries and companies all try to diversify their chokepoint risk.

Appendix 1

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The Expanding Frontier of Agentic AI: A User's View

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The most important change in AI today may not be the arrival of a single new frontier model, but the speed at which frontier-like capability is becoming cheaper. My own use of AI agents since March suggests the shift is already visible: models that once looked like expensive high-end choices are being matched by cheaper alternatives within months. This report looks at that expanding frontier and what it means for adoption.

Since I started using AI agents in March, trying new models has become something of a hobby—and partly a necessity. I could not rely on the most advanced models to power my OpenClaw Bots and other tools, not only because access is not always guaranteed, but also because they are extremely expensive for agentic usage. I therefore spent the past few months switching among more accessible and affordable alternatives, many of them models developed by Chinese labs.

The effect of each switch was often not obvious. A model might seem more reliable or require less instruction, but the improvement was difficult to separate from chance. Over a longer period, however, the change became unmistakable. My AI agents could complete the same tasks with fewer prompts, use tools more effectively and travel further from a simple instruction before requiring intervention.

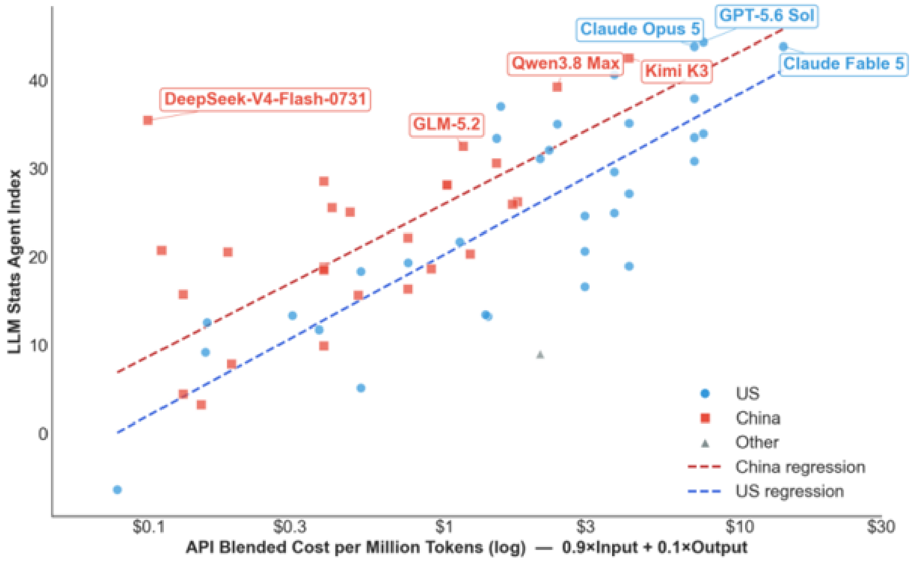
That said, two recent model releases did feel different to me. Both Kimi-K3 and DeepSeek-V4-Flash-0731 seemed notably more powerful than the previous alternatives I had been using. Was this merely a subjective experience, or had the underlying models genuinely improved? This prompted me to do some databased analytical research.

Measuring agentic intelligence against cost

The research I did was a simple plot of popular models along two dimensions: capability and cost. The horizontal axis is API cost per million tokens on a log scale, using a simple blend of 90% input tokens and 10% output tokens. The vertical axis measures model capability. There are multiple possible measures, but I chose the Agent Index from LLM Stats for two reasons: first, I care more about agentic capability than other dimensions such as coding or writing because of my use case; second, LLM Stats offers a composite measure based on benchmarks involving tool use, multi-step execution, web search, terminal operation, software engineering and other agentic tasks.

The broad pattern is intuitive: more capable models tend to be more expensive. US models are shown in blue, and Chinese models in red in the chart. A clear upward correlation can be observed across all the dots. The difference is that US models occupy much of the upper-right part of the chart, while Chinese models are more densely represented at lower price points. The regression lines capture this difference: both groups exhibit a positive relationship between cost and agentic performance, but the US cluster generally extends further into the highcost, high-capability region.

US and Chinese AI Models copared on cost and Agentic capability



Source: Deutsche Bank Research, LLM Stats

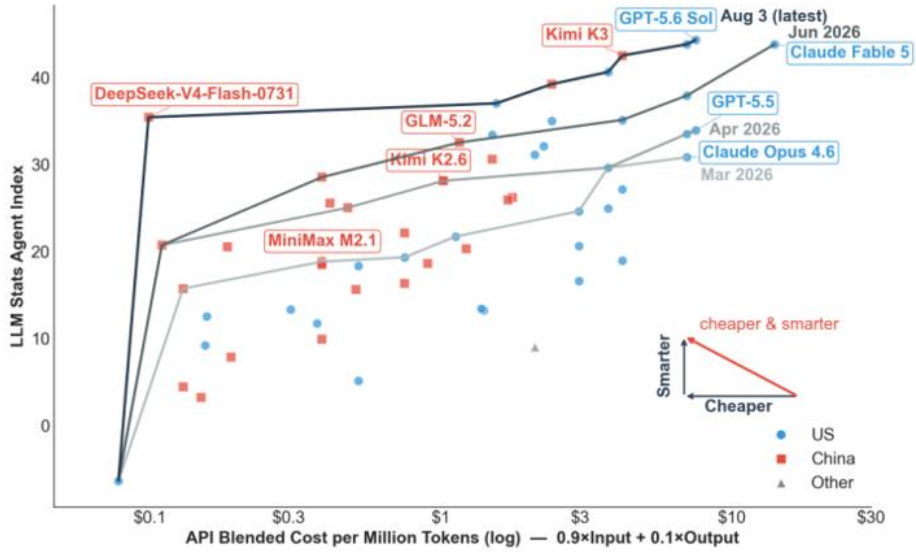
A frontier moving rapidly

The more useful concept for model selection—at least for me as an economist— is the Pareto frontier. A model is Pareto optimal if no cheaper model has a higher score. Connecting all Pareto optimal models gives the Pareto frontier: the boundary of best available model performance at each cost level. A model below that frontier may still suit a particular task, but on these two dimensions it is dominated by another choice. The chart below shows the Pareto frontier for all models in my dataset.

For example, at the top of the current frontier, GPT-5.6 Sol has the highest Agent Index in the dataset, at 44.3, with a blended cost of $7.50 per million tokens (the dot next to it is Opus 5). This is just too expensive for me. If I want to reduce cost, the next best choice according to the chart would be Kimi-K3: with only a modest loss in capability, it offers a meaningful cost saving and could be a good choice for slightly less complex tasks. But it is still too expensive for simple tasks, for which I would go even further down the line to maybe DeepSeek-V4-Flash.

But the analysis did not stop there. The chart also shows how the Pareto frontier has shifted as new models became available. By including only models released by each cutoff date — March 31, April 30, June 30 and August 3 — I was able to reconstruct earlier frontiers and compare them with the latest one.

Expanding Pareto Frontier: Agentic AI is becoming cheaper and smarter at speed



Source: Deutsche Bank Research, LLM Stats

The speed of Pareto improvement is striking. Every month, the Pareto frontier moves upward and left by a large margin. This means users can achieve better performance at the same cost, or cut spending significantly while maintaining similar performance. For example, GLM 5.2 on the June frontier has almost the same score (32.5) as Opus 4.7 (33.5) on the April frontier, while its blended cost of $1.16 is far below Opus 4.7's $7. In merely two months, a user could potentially reduce token cost by more than 80% without losing performance.

The latest frontier line makes the point even more clearly. As of today, almost all models on the latest Pareto frontier have exceeded GPT-5.5's score. In other words, **what sat on the high-cost frontier in April is now available across the cost spectrum.**

Chinese AI models are catching up

It would be easy to summarize the chart by saying that US models remain ahead… At the absolute frontier, that is still true: GPT-5.6 Sol holds the highest Agent Index in the dataset. But that reading misses the more consequential competitive dynamic.

…but Chinese models' advancement from the lower-left is also remarkable. In March, when I first started using AI agents, the Pareto frontier was dominated by US models. Kimi K2.6 moved into the middle by April; GLM-5.2 approached the performance of recent leaders by June; and Kimi K3 narrowed the gap near the top in July. Kimi K3, being the largest (2.8T parameter) and most costly openweight model to date, shows that Chinese models are no longer competing only on price; they are increasingly present near the capability frontier as well

The DeepSeek Outlier

What strikes me most is how the latest DeepSeek release altered the shape of the chart. DeepSeek-V4-Flash-0731, released last week, was a large improvement over the earlier low-cost version included in the April frontier. The wearlier model was already inexpensive, with an Agent Index of 20.7 at $0.11 per million tokens. The July release has effectively the same cost, but its Agent Index rises sharply to 35.4. It moves DeepSeek from the low-cost edge of the chart into the performance range of much more expensive frontier models.

DeepSeek demonstrates how much potential remains for AI models to improve. DeepSeek-V4-Flash is far smaller than models near the current frontier. At 284B parameters in total (13B active under a Mixture-of-Experts architecture), its agentic capabilities outperformed much larger models such as the 744B GLM-5.2 and the 1.6T DeepSeek-V4-Pro. The improvement was achieved not through increasing model size (making it more knowledgeable) but through post-training (teaching it to better handle tasks). The contrast illustrates that good agent performance does not require the largest model; architecture and post-training can move the effective frontier more than parameter count alone would suggest.

The Jevons Paradox is Working on Me

Another interesting fact: the steep decline in AI model costs did not bring me any savings. The opposite is happening—my AI bills have surged. My total spending on AI was effectively zero in February before I started using AI agents. Then I started using OpenClaw and made my first token subscription plan at $20/month. A few months later, my recurring spending on AI is now reaching $100-200/month. My costs increased because I now use AI agents to do many more things and must subscribe to new plans as tokens run out under existing ones. The increase also includes cloud storage and an $800 Mac Mini to host my "little lobsters" (OpenClaw Bots).

This is a familiar economic mechanism: the Jevons paradox. Improvements in efficiency did not reduce consumption; rather, they encouraged wider use and led to an increase in total usage and total spending. As model capability improved while token cost dropped, I expanded the use of my AI agents to more daily tasks—tracking headline news and specific topics I am interested in, summarizing market movements, researching academic papers, managing expenses, making travel plans and testing ad-hoc ideas. Why use a chatbot if you have dedicated AI agents that know your background and remember the conversation history?

The question is therefore not simply which model will hold the highest score next month. It is what happens when millions of users and businesses can afford to keep capable agents running for longer, on more tasks, with less concern about every token they consume. If the cost of agentic intelligence continues to fall while its capability rises, will efficiency save tokens—or will Jevons' paradox help trigger an explosion in real-world AI use?

My verdict: the expansion of the AI Pareto frontier is benefiting users and the broader AI ecosystem. As US and Chinese labs compete, premium frontier models push the performance ceiling higher, while efficient lower-cost models make agentic workflows economical at scale. Together, they allow users to match each model to the value and difficulty of the task, rather than accept a single price-performance trade-off. Competition is accelerating the race, while the global circulation of research ideas, open weights and practical techniques is helping improvements spread quickly across the field. **The analysis made me smore confident that the future of agentic AI is arriving faster than many expected.**

Appendix 1

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Situational awareness - do we have it?

August 3, 2026

Executive summary

There are two events in the last few weeks that look like non-events. The Situational Awareness hedge fund unwound a $16bn book of leveraged AI trades with barely a ripple beyond a few days of volatility. Meanwhile, the Kospi fell more than 20% in 48 hours and then rallied 25% off the low — finishing the week almost flat. Taken individually, both look like markets doing their job: absorbing shocks and moving on.

We think that read is wrong, or at least incomplete. Both events were symptoms of the same underlying condition: leverage in ETFs, margin accounts and hedge fund books that was encouraged by more than a decade of low and stable rates. Indeed, reports that over 3% of Korea's adult population received a margin call over the last two weeks is deeply concerning if true.

This piece argues that 'small vol' events are going to affect markets with an increased frequency. That comes as the backdrop of normalised interest rates and geopolitical unrest has been augmented by a Fed that has moved away from the forward guidance that reduced some volatility in the past.

We also worry that the risks in household debt (and investment accounts on margin) is being ignored as the focus stays on the risks in growing sovereign debt. If this is right, 'small vol' episodes are likely to become more frequent and potentially have larger consequences.

In the end, we worry that investors, corporates and policymakers are focused on avoiding the next big financial crisis and, in doing so, may be underestimating the risk in small events that can cascade.

A tale of two 'small vol' events

Markets have spent the last few years braced for another 2008-style event. That vigilance has a blind spot. It trains attention on the big, obvious risks and away from the smaller, everyday ones that can cascade in the right circumstances.

The implosion of the Situational Awareness hedge fund is the first case in point. The fund sold its $16bn public equity book after highly leveraged trades on AI companies went wrong. Markets absorbed the unwind with only a brief bout of volatility — no contagion, few headlines beyond the financial press.

The second was in Korea. At one point last week, the Kospi fell over 20% in barely 48 hours, before staging a 25% rebound from the low. The headline index ended the week close to flat and broader markets have shrugged it off. But a flat index conceals an extraordinary amount of forced selling and buying beneath the surface, much driven by margin calls on leveraged retail positions.

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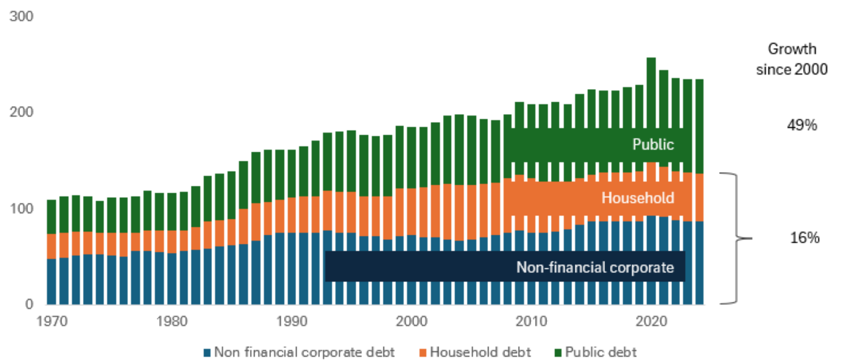
Why ‘flat on the week’ is the wrong takeaway

An index that round-trips 45 percentage points in days is not the same thing, economically, as one that simply sits still. Leveraged and margined positions are typically closed out mechanically and irreversibly on the way down, regardless of what happens on the way back up. The P&L may net out at the index level; the wealth destruction for individual, often first-time, leveraged investors does not.

We believe the potential for household leverage to cause problems is underestimated because the aggregate figures look okay. When we show the following chart, clients gravitate to the rise in sovereign debt, not the plateau in household leverage.

Figure 1: Household debt has lost the spotlight to public debt since 2000

Global debt as a % of global GDP

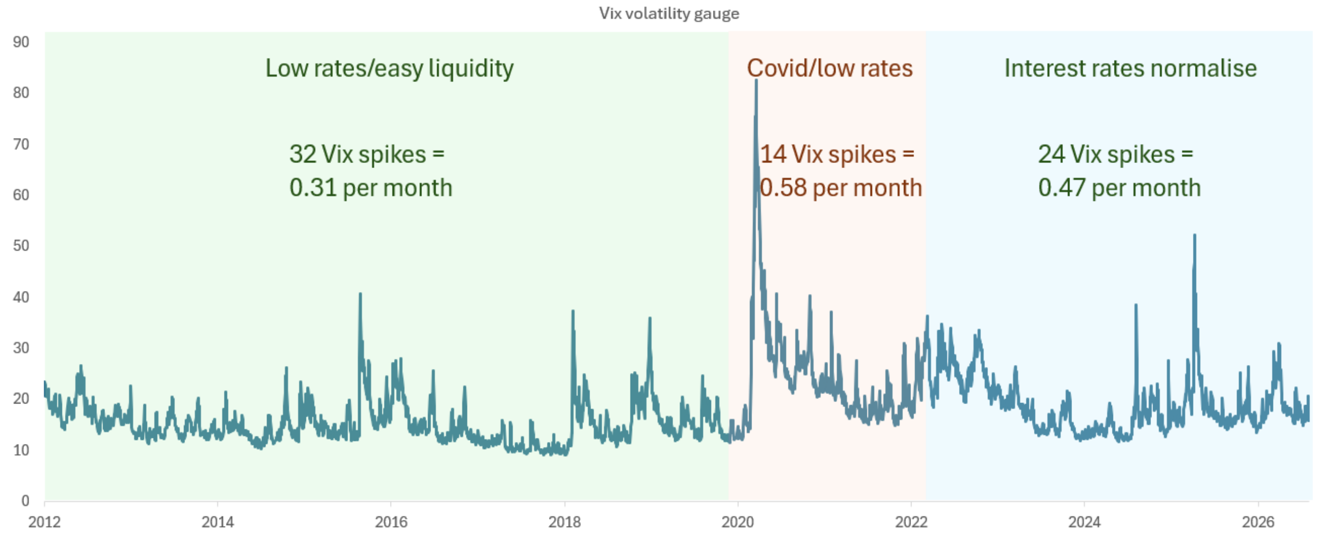


Source: IMF, Deutsche Bank Research

Small vol events are becoming more frequent

The following chart is created using our own judgement. But it shows that, by our reckoning, the number of spikes in the Vix since 2022 has been materially higher over the last few years than during last decade. The last four years are an important comparison period as this is when interest rates began to normalise.

Figure 2: A greater frequency of 'small vol' events appears to be the norm, at least for now



Source: Factset, Deutsche Bank Research

Is the Fed trying to bring volatility back on purpose?

One explanation for why policymakers may be tolerating episodes like these comes from Rob Armstrong (a former colleague) here. He advanced a theory about Fed chair Kevin Warsh that essentially says, his withdrawal of forward guidance is a deliberate attempt to reintroduce 'uncertainty premia' into markets that have used lower volatility to take on more leverage than they otherwise would. The idea is that "Too little everyday volatility ultimately leads to financial crises."

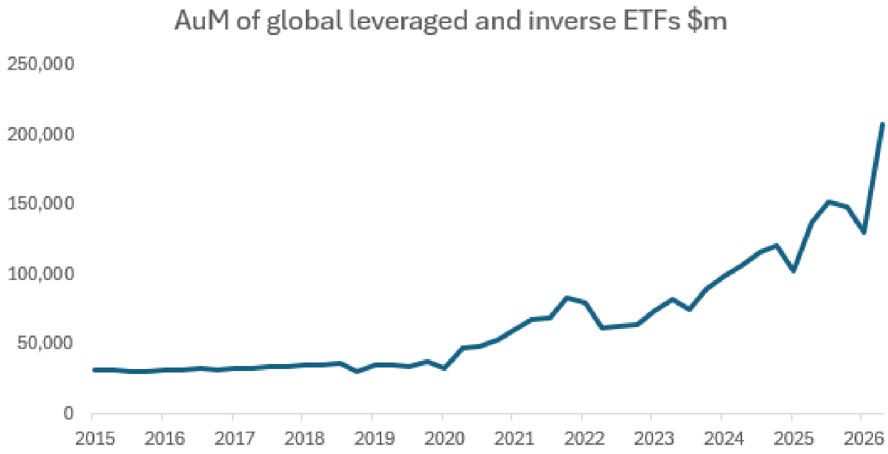
The mechanism, on this reading, is that forward guidance suppresses everyday volatility — and stable prices and calm markets encourage more borrowing, both by governments and by market participants, than would occur under normal price discovery. Warsh cannot say this outright without spooking markets, hence the vaguer 'referee' rhetoric — but the policy actions, in Armstrong's view, point the same way.

If policymakers genuinely believe more day-to-day volatility is healthy, investors should expect more 'small vol' episodes like Situational Awareness and the Kospi swing and should stop treating each one as isolated, ring-fenced events.

Leveraged ETFs and margin debt: Should we be afraid?

Leveraged and inverse ETFs, and the margin debt that often sits alongside them, are growing quickly across many major markets. They are a key mechanism that turns an ordinary correction into a cascade because they force mechanical, non-discretionary selling as prices move.

Figure 3: The semiconductor and broader tech rally has fuelled the popularity of leveraged ETFs



Source: Bloomberg Finance L.P., Deutsche Bank Research

In Korea specifically, the launch of single-stock leveraged ETFs tracking Samsung Electronics and SK Hynix in April turned two already-dominant, trillion-dollar-plus stocks into vehicles for retail leverage almost overnight. Retail investors accounted for a large proportion of holders. When the underlying stocks fell, those products amplified the move.

The risk is not confined to Korea. The same products are growing in the US and Europe. A market that is many times larger, and far more directly connected to consumer spending and credit availability, having its own 'Kospi moment' is a materially bigger problem for systemic risk than a niche hedge fund unwinding.

A footnote: why the name 'Situational Awareness' went viral

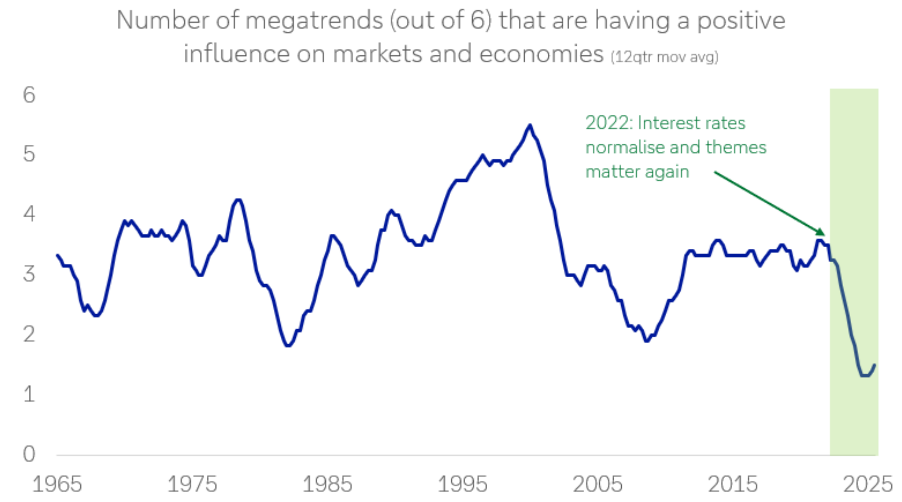
The fund's name borrowed from Leopold Aschenbrenner's 165-page 2024 essay, which argued AGI was plausible by 2027 and sketched a path from massive compute build-outs to an 'intelligence explosion' and eventual government-led consolidation of frontier AI ('The Project').

Part of a developing medium-term trend

Our megatrend framework tracks six long-run forces (technology, sovereign deficits, geopolitics & globalisation, domestic politics & social discontent, demography, and energy) and combines them into a single indicator of how supportive the backdrop is for markets and economies.

Overall, the number of megatrends working in favour of markets and economies has fallen to a level last seen only twice before — during the 1970s oil crises and around the 2008 financial crisis. Both were periods marked by exactly this kind of 'small vol' pattern: idiosyncratic-looking events that, in hindsight, were symptoms of a more fragile underlying system.

Figure 4: Prior periods of megatrends being this deeply negative have resulted in the greater frequency of volatility events



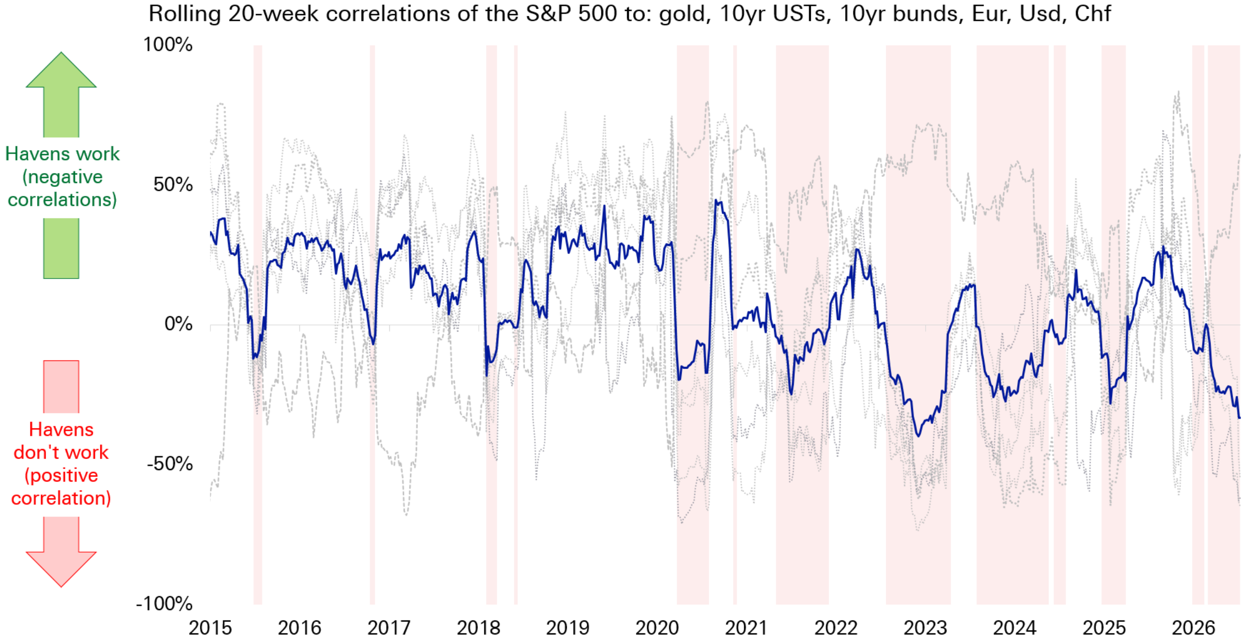
Source: Deutsche Bank Research

The single most important negative megatrend is sovereign deficits. Developed-market governments are refinancing and issuing new debt at rates well above those of the last decade. At the same time that our model flags unpredictable sovereign debt sell-offs, particularly those triggered by exogenous shocks and investor emotion, as a growing risk.

Havens are not there to catch you

Ordinarily, investors would look to traditional havens (gold, the dollar, the Swiss franc, the yen, US treasuries and bunds) to cushion volatility. However, our haven indicator shows that, since 2020, this combination of assets has failed to hedge risk (as proxied by the S&P 500) particularly during more shock events, including covid, the 2022 rate hikes, the 2025 tariff shock and the 2026 Iran war.

Figure 5: Traditional haven assets have not worked in the 2020s, particularly during shock events and periods of volatility



Source: Bloomberg Finance L.P., Deutsche Bank Research

If havens keep failing when they are needed most, the cushioning effect investors implicitly rely on to keep 'small vol' small is weaker than it used to be. That raises the odds that future 'small vol' episodes may cascade into something more.

What if this happened in the US or Europe?

It is tempting to treat the Kospi swing as a Korea-specific story — a function of retail culture, single-stock products and two dominant, correlated semiconductor names. But if a similar episode plays out in US or European, the results could be systemic.

The transmission mechanism would not be limited to portfolio losses. It would run through consumer spending, as households with margin-called leveraged positions cut back; through credit, as brokers and lenders reassess collateral and risk; and through confidence, as retail participation reverses. A leveraged retail shock in the US would be a materially larger systemic risk than a single hedge fund unwind.

Another example

Silicon Valley Bank is another example. On paper, SVB's failure should have been a contained, idiosyncratic event — a mid-sized bank with a concentrated depositor base and a duration mismatch. In practice, it triggered enough of a confidence panic that policymakers felt compelled to effectively guarantee all deposits across the banking system, a far larger intervention than the initial problem seemed to warrant.

The lesson is not that every small event becomes systemic but the ones that do are rarely obvious in advance. And when there is an increased frequency of these events in a market with more leverage, less reliable havens, and policymakers newly tolerant of everyday volatility, there is an increased risk that 'small vol' events may catalyse larger problems.

Conclusion: are we situationally aware?

The market's instinct is to file Situational Awareness and the Kospi swing under 'noise' because neither, so far, has cascaded into a systemic event. That instinct may be dangerous, particularly if policymakers are willing to tolerate more everyday volatility.

Given our megatrend model indicates that the supportive backdrop for markets is near a multi-decade low, 'small vol' episodes can potentially be used as an indicator of how stress in the financial system is quietly building up.

There are plenty of other parts of the market where 'small vol' events may catalyse problems in areas that were built up with leverage during the periods of easy money. Just two examples are:

- Private equity: a large amount of TMT portfolio companies bought during the era of near-zero rates remain on PE firms' books, held longer than usual as exit markets stayed shut. A reckoning may still be ahead.
- Corporates: many listed companies still carry underperforming divisions bought during debt-fuelled M&A booms last decade. As a result, they have lower asset turnover metrics, a legacy of a decade in which cheap financing incentivised growth in revenue over earnings.

Appendix: Why Situational Awareness went viral

The original essay by Leopold Aschenbrenner in 2024 was 165 pages long. Below is a bullet point summary (generated by AI).

The Path to AGI by 2027: The document argues that Artificial General Intelligence (AGI) is strikingly plausible by 2027. This prediction is based on extrapolating "Order of Magnitude" (OOM) trendlines in three areas: physical compute scaling, algorithmic efficiencies, and "unhobbling" gains (transforming models from chatbots into active agents).

Massive Industrial Mobilisation: AI is shifting from a software boom to a massive industrial process. Leading AI labs will build training clusters costing $100s of billions by 2028 and over $1 trillion by 2030. By the end of the decade, these clusters could require power equivalent to 20% of total U.S. electricity production.

The Intelligence Explosion: Once AGI is achieved, it will likely be used to automate AI research itself. Because digital researchers can run in millions of copies and at much higher speeds than humans, a decade of algorithmic progress could be compressed into a single year, leading rapidly from AGI to superintelligence.

The Security Crisis: Current security at leading AI labs is described as "Swiss cheese," leaving key algorithmic secrets and model weights vulnerable to stateactor theft, particularly by the CCP. The author warns that leaking these secrets in the next 12-24 months would be a catastrophic national security failure.

Superalignment Challenges: Reliably controlling AI systems that are vastly smarter than humans is an unsolved technical problem. Current alignment techniques (like Reinforcement Learning from Human Feedback) will break down because humans cannot supervise behaviour they do not understand. Solving this will require using AI to help automate alignment research.

"The Project" (U.S. Government Intervention): Given the decisive military and economic advantage of superintelligence, private startups cannot handle it alone. By 2027 or 2028, the U.S. government is expected to initiate a nationalized AGI effort—referred to as "The Project"—to consolidate resources, secure model weights, and ensure the free world prevails over authoritarian powers.

Geopolitical Stakes: Superintelligence will provide a military edge comparable to nuclear weapons. A lead of even a few months could be decisive, making a healthy lead for democratic allies essential for safety and global stability.

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Global events matter, but don't, but do

July 30, 2026

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Executive summary

There is a common assumption that markets look through global events such as war and geopolitics. But the effects of these and other events extend far more broadly into investor portfolios than just the headline indices. Our latest data indicates just how much most people oversimplify the reaction of investors and markets to global events.

Indeed, we find that investors do not look through global events. On the contrary, they are highly polarised about them. On our analysis, the vast majority of investors think global events will either be a "risk" or an "opportunity" for markets. That is very different from almost all other broad themes where a higher proportion of people simply think there is "no effect".

Our dbDataInsights team asked 500 people from the US and UK if and why they think global events affect their investment portfolios and what they are doing about it. We then used AI to analyse all this unstructured data. In this piece, we compare the results to specific datapoints that indicate investors are actively reshaping their portfolios to either capture opportunities or hedge the risks that they see in global events.

In the end, we see that positive and negative views on global events can coexist. The negative respondents focus on the immediate effects: oil prices rise, inflation rises, markets fall. The positive respondents focus on longer-term effects: markets adapt, companies innovate, energy systems diversify, and portfolios positioned in the right sectors benefit. Both views may be internally coherent; they simply operate on different time horizons.

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Why global events are making a comeback

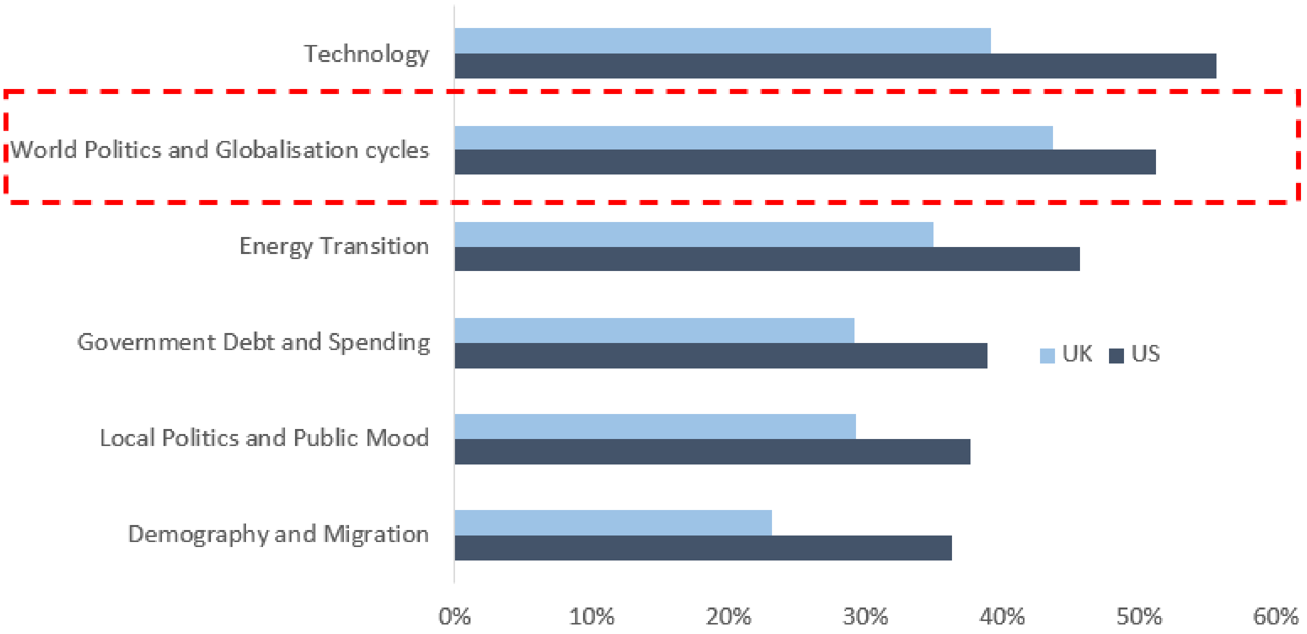
Global events are more than just war and geopolitics. They extend to incidents that affect energy prices, inflation, fiscal sustainability, and technology. Indeed, these forces are some of the strongest that affect global economies and we discuss them in detail in our piece on global megatrends.

Our thesis is that, for much of the post-financial-crisis period, low interest rates drowned out the other effects of global events. Low rates raised valuations, pushed investors into risk assets, and made it easy to explain market moves through central bank policy. Now that rates have normalised, global events are more visible as market drivers again.

Look through or not, investors are changing their portfolios

Our data indicates that investors are reshaping their portfolios as a result of global events. These investors view them not as merely one-off shocks but things that are becoming persistent forces.

Figure 1: Over the last three years, have the following thematic megatrends resulted in you making changes to your investment portfolio?



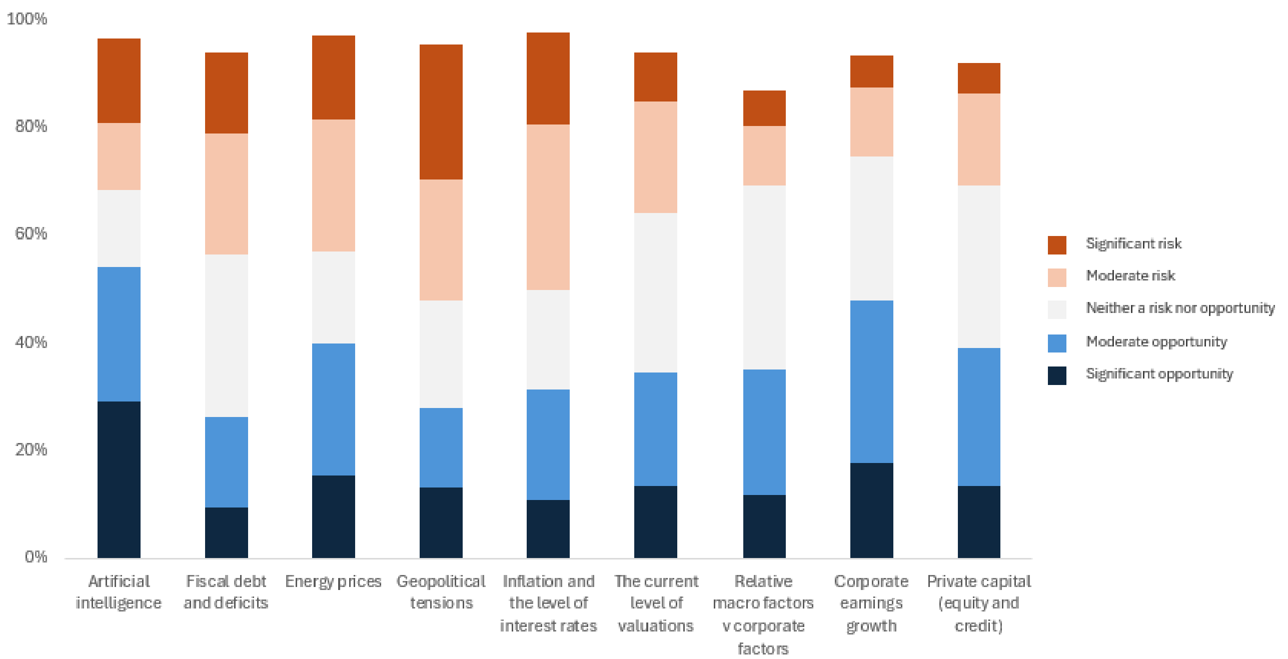
Source: Deutsche Bank dbDataInsights

The investor base is therefore not split between people who "understand" global events and people who do not. Rather, it is split between investors who emphasise different time horizons and different asset exposures. A broad equity investor may fear a Middle East shock because it threatens inflation and sentiment. An energy, gold, or defence investor may welcome the same shock because their holdings could benefit. A long-term diversified investor may view the event as noise.

Investors are highly polarised about global events

Retail investors are not passive observers as many may believe. The following table shows that our surveyed investors have strong beliefs both ways about global events and their impact on portfolios. Most importantly, when we look at geopolitics, energy prices, and inflation, the proportion of people who think there is no effect is very low compared with the other themes.

Figure 2: For each investment theme, is it a risk or an opportunity for investors in 2026? (US)



Source: Deutsche Bank dbDataInsights

Aside from the shock effect, it is understandable that the same global event can produce very different portfolio expectations. A war in the Middle East can be viewed as a risk because it raises oil prices, hurts consumers, and depresses equities. But it can also be viewed as an opportunity if an investor owns energy stocks, gold, defence shares, or believes that markets will rebound after the shock. Indeed, in our survey, several positive-impact respondents explicitly mentioned these assets and actions.

Energy is a good example. Short-term energy disruption is painful for consumers and companies. Yet, energy disruption can have positive medium-term effects if it pushes economies toward greater energy security, diversification of supply, and lower energy intensity. Indeed, since 1990 the amount of GDP generated per kilogram of oil equivalent has quadrupled in terms of purchasing power parity. In other words, economies have become much more energy efficient over time which helps explain why the current closure of the Strait of Hormuz has not yet had the disastrous effects of the oil shortages during the 1970s.

This is consistent with our megatrend framework which shows geopolitics and globalisation do not have simple linear impacts on economies. Trade in goods can be reorganised through supply chains, and trade in ideas can continue to accelerate through connectivity. In other words, geopolitical deterioration is not always a pure negative for markets. It changes the winners and losers or increases dispersion.

The most important topics

The following table shows the top 3 themes that were observed in our survey of why people think global events are either positive, negative, or neutral for their portfolios.

Figure 3: Top 3 types of global events that matter to people

Global events that make people more positive, negative, or neutral about their portfolio			
US	Negative	War and geopolitical conflict, especially Iran, Middle East, Ukraine/Russia and escalation risk	28%
US	Negative	Inflation and cost-of-living pressure reducing returns and company earnings	14%
US	Negative	Oil, gas and energy price shocks feeding through to markets	13%
UK	Negative	War and international conflict, especially Iran, Middle East, Ukraine/Russia and wider unrest	26%
UK	Negative	Inflation, rising prices and higher living or business costs	20%
UK	Negative	Energy, oil and fuel price shocks, including Middle East supply disruption	17%
US	No impact	Diversification across assets or holdings expected to absorb shocks	25%
US	No impact	Long-term investing, with short-term events expected to smooth out over time	18%
US	No impact	Low-risk, fixed income, cash-like or insulated investments	18%
UK	No impact	Long-term investing, with market ups and downs expected to smooth out	25%
UK	No impact	Uncertainty, lack of knowledge or unclear rationale	22%
UK	No impact	Diversification or resilient portfolio construction	18%
US	Positive	Global events creating growth opportunities, new markets or higher returns	20%
US	Positive	Technology, AI, digital transformation, healthcare, renewable energy or innovation	16%
US	Positive	Economic recovery, stabilisation and improving market confidence	15%
UK	Positive	Economic recovery, stronger markets and improving investor confidence	20%
UK	Positive	General optimism or confidence that things will improve	16%
UK	Positive	Global events creating investment opportunities, new markets or growth sectors	15%

Source: Deutsche Bank Research

Energy prices are the everyday face of geopolitics

Energy prices are the mechanism through which many investors experience global events most directly. In the written responses, distant conflicts become financially relevant because they raise oil, gas, petrol, utilities, shipping costs, food prices, and inflation. This is especially visible in the UK negative responses, where energy costs and inflation appear repeatedly.

Energy prices are viewed as a risk by 40% of the US investors surveyed and 55% of the UK investors. They are also viewed as an opportunity by 40% of the US investors and 29% of UK the investors. The polarisation here is obvious, and the country difference matters. The US has a large domestic energy sector, meaning some investors may see higher energy prices as positive for certain holdings. The UK investor base, by contrast, appears more likely to interpret higher energy prices through the household cost-of-living and company-cost channels.

Why some investors think global events do not matter either way

These investors are not necessarily saying that global events do not matter for the world. They are saying global events do not matter enough for their portfolios.

The main reasons both the US and UK investors in our survey cite are diversification, long-term investing, defensive holdings, local exposure, and financial discipline. They also frequently think that portfolio outcomes are driven more by company fundamentals, dividends, valuations, financial advice, local exposure, or long-term compounding, than geopolitical events. In the written responses, this often appears as "markets go up and down," "over time they smooth out," "my portfolio is diversified," or "I invest for the long term."

There is also a behavioural component. Some investors may choose "no impact" because they do not feel confident predicting global events. Several responses in the 'no impact' columns are uncertain, vague, or non-committal. This does not mean respondents believe global events are irrelevant. It may mean they do not know how to connect those events to their specific investments.

The "no impact" camp is therefore divided into two groups. The first is strategically insulated: diversified, long-term, defensive, or locally exposed. The second is cognitively uncertain: unsure, disengaged, or unable to explain the link.

Why some investors think global events will hurt their portfolios

The 'negative impact' responses are dominated by concerns about geopolitics. In the US, the most important theme is war and geopolitical conflict, particularly Iran, the Middle East, Ukraine/Russia, and escalation risk. In the UK, negative responses are similar but are even more explicitly tied to the inflationary consequences of global events. They frequently mention energy prices, oil, gas, the cost of living, and the way foreign conflicts can feed into domestic household and business costs.

When investors are negative about global events, their written responses suggest that many see a clear macro chain:

1. Global conflict disrupts energy, trade, or confidence.
2. Energy and commodity prices rise.
3. Inflation and interest-rate pressure increase.
4. Company margins and consumer spending weaken.
5. Stock markets fall and portfolio values decline.

This chain appears again and again in the responses, even when written informally. Investors may not use terms such as "supply shock," "margin compression," or "risk premium," but they are effectively describing those concepts.

The UK-US difference is notable. Our surveyed US investors appear somewhat more willing to see opportunities within global volatility, while the UK investors are more likely to view the same themes through the lens of inflation, energy dependence, and broader economic vulnerability. That may explain why "World Politics and Globalisation cycles" is the highest-ranked cause of portfolio change among UK respondents.

Political uncertainty also has a strong presence in the written responses. In the US, this focusses on President Trump, tariffs, foreign policy, and the perceived unpredictability in government. In the UK, people also mention Trump and US policy, but usually as part of a wider concern about global instability. This implies that investors do not separate domestic politics from geopolitics as cleanly as analysts often do. Leadership, war, tariffs, inflation, and markets are therefore perceived as part of one connected risk cluster.

Why some investors think global events will help their portfolios

The positive camp contains three broad types of investors. First, the optimists who believe technology and growth will dominate. Second, the sector opportunists who own assets that benefit from disruption. Third, the tactical buyers who treat volatility as a chance to buy assets cheaply.

Notably, many investors say they apply a clear investment logic. Essentially, 'disruption creates opportunity'. They cite the effect of global events on an economic recovery, technology, AI, energy transition, lower prices after market sell-offs, defence, oil, gold, emerging markets, and international trade recovery.

The most important positive theme in the US responses is growth opportunity: global events can open new markets, increase demand, create new investment opportunities, and raise returns. Technology and AI are also central. Many respondents expect AI, digital transformation, health care, renewable energy, and innovation-led sectors to do well.

This US-UK divide appears again in the written responses. US positive respondents are frequently enthusiastic about AI, tech, innovation, market growth, and sector opportunities. UK respondents are also positive on technology and innovation, but their positive responses more often include general optimism, economic recovery, and rebounds after conflict.

There is also an opportunistic trading mindset. Some respondents say market dips allow them to buy, or that volatility creates opportunities.

Indirect effects that are implicit in investor logic

The most concerning megatrend in the world today is fiscal debt and deficits – and it is an important part of the broader story about global events. Fiscal issues are less prominent in the written responses from investors compared with war, oil, inflation, and President Trump. Yet the implicit themes run through many identified factors. Indeed, 38% of US and UK respondents view fiscal debt and deficits as a risk. At the same time, around a quarter of people in both countries view them as an opportunity.

So, although investors talk frequently about inflation, rates, taxes, government borrowing, and weaker growth, they do not always connect those concerns explicitly to sovereign deficits. But fiscal anxiety appears indirectly embedded in worries about inflation, interest rates, debt, taxes, and government policy.

Government debt and spending has already prompted portfolio changes by 39% of the US investors and 29% of the UK investors over the last three years. That is lower than technology, world politics, and energy transition, but still significant. It suggests that fiscal concerns are not absent, they are simply less emotionally vivid than war or energy prices.

Thus, investors who are positive on AI may be implicitly assuming that technology can overpower fiscal and geopolitical headwinds. Investors who are negative on global events may be implicitly assuming that debt, conflict, inflation, and policy instability will overpower technological progress. The debate is not simply about whether global events matter. It is about which global force dominates.

Conclusion: The polarisation of opinion on global events means we should brace for volatility and dispersion

In the end, the surveyed investors overwhelming think global events matter because they reach into their portfolio through inflation, oil, politics, growth, and technology. Investors think the world does not matter when they believe their portfolio construction, time horizon, or asset selection can neutralise those forces.

The investor base is therefore not split between people who “understand” global events and people who do not. Rather, it is split between investors who emphasise different time horizons and different asset exposures.

Cultural differences are important. The surveyed US investors are most likely to have changed portfolios because of technology, followed by world politics, while the UK investors are most likely to have changed portfolios because of world politics and globalisation cycles, and then technology.

This suggests that US investors are more likely to frame the future around innovation and technology while UK investors see investing more through the lens of global political cycles.

In investment terms, this means global events should be understood less as a single directional call on equities and more as a source of dispersion. In other words, they just change the relative winners and losers.

Appendix 1

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Deutsche Bank
Research Institute

What a "Strong and Rich Japan" could mean

July 24, 2026

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Executive summary

- Japan could be at the precipice of a very significant shift in fiscal and industrial policy, with echoes of the Meiji Restoration. Japan's new economic blueprint seeks to put the country on a path of investment-driven growth to meet the challenges of a new era.
- Japan is not alone in its bid to prioritize industry and rearm, but given a higher starting point for government debt, it will have to find room to spend while maintaining fiscal sustainability. This will inform Japan's approach in very significant ways.
- First, Japan is likely to remain focused on supporting nominal GDP growth, aiming to keep it above government funding costs to create deficit space. Second, Japan will look to actively mobilize domestic savings to fund investments and manage yield pressure. Each of these shifts could have implications for the currency.
- Japan's excess savings have arguably been kept in two places: in cash and abroad. The government is looking to get both moving. Households have around 50% of their $15tn of savings in cash and deposits. Meanwhile, GPIF has 50% of its $1.8tn portfolio abroad. If industry, energy, defence need to come home, so might the capital to fund them.
- Over the past few months, Japanese policy makers have been very focused on stabilizing the currency. However, if fiscal capacity is becoming the most important policy criteria, incentives could shift from FX to yield management.
- The ultimate impact on USD/JPY will depend on the levers the government chooses to pull to manage yields. If GPIF is mandated to bring money back into domestic assets, this could be very bullish for the JPY. However, if the BoJ is coopted to support bonds through renewed JGB purchases, this could be very negative for the JPY.
- Suppressing yield volatility could come alongside bigger moves in FX from here, depending on the approach the government takes. This favours higher JPY vol.

Echoes of Meiji

Japan could be at the precipice of a very significant shift in fiscal and industrial policy, with echoes of the Meiji Restoration. In 1868, pre-modern Japan was staring at the threat of foreign involvement and colonization, having seen the Opium Wars unfold in China. Japan had signed unequal trade treaties that opened Japanese ports to free trade imperialism, prevented Japan from setting tariffs, and exempted foreigners from local laws. Japan's response changed history: under the Meiji Restoration, Japan returned rule to the emperor, strengthened the state, and pursued extremely rapid industrialization.

One could argue that Japan again faces existential challenges given fractures in globalization, intensifying great power competition, insecurity around energy and critical minerals, the need to participate in AI innovation, and external dependence for defence. Japan's answer under PM Takaichi: to return to its roots as an industrial nation driven by state and fiscal policy. The slogan of the 1868 Meiji Restoration was "Fukoku Kyohei" which translates into "Enrich the Country, Strengthen the Armed Forces" or more simply: rich country, strong army. It is perhaps telling that Takaichi's government is introducing an economic blueprint that calls for a "Strong and Rich Japan". (Nikkei Asia).

Invest, invest, invest

Japan's Cabinet has now passed the Honebuto, or the government's new Basic Policy on Economic and Fiscal Management. The document calls for moving beyond "conventional thinking", "putting an end to the trend of underinvestment", and "responding to the era of intense global industrial policy competition". The cornerstone of Japan's proposed policy shift seems to be Y370trillion of public-private partnership investments with the government having identified 17 key strategic areas from AI and quantum, to defence, aviation, shipbuilding, critical minerals, and others. These investments are seen as crucial to boosting potential growth. Indeed, the decline in Japan's potential growth rate has not principally been about demographics as is commonly assumed, but about the collapse in the capital stock (Figure 1). Japan went through decades of very low investment after the 1990 asset bubble burst. It now wants to return to being an investment-driven economy.

Finding the funding: two big shifts

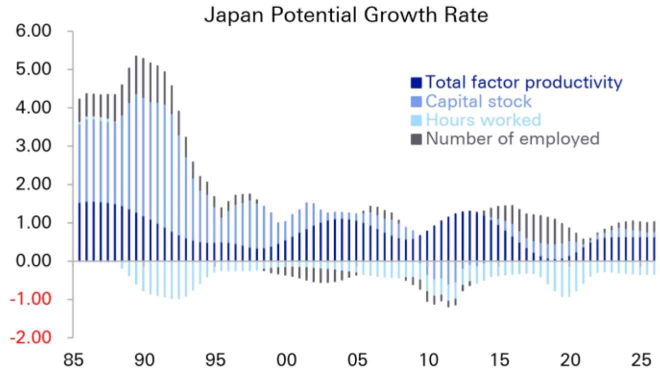
Japan is not alone in its bid to reindustrialize and rearm, but compared to peers, Japan arguably has less fiscal space. Japan's debt to GDP hovers above 200%, while Germany and Korea are under 50%, making it easier for them to pursue significant expansion in their fiscal deficits. While Germany is moving from austerity to expansion, Japan will have to find spending room while continuing to chart a fiscally sustainable path. This will hugely inform Japan's policy approach. There are two important shifts Japan appears to be making to enable fiscal spending.

First, changing the fiscal target. The Honebuto has confirmed a shift in Japan's core fiscal target away from the primary balance. The primary balance will now no longer be a mechanical target; it will be managed over many years, and be allowed to deteriorate temporarily as required. The new fiscal target is now the "stable reduction of the total debt-to-GDP ratio for national and local governments." The second notable and enabling shift is the pursuit of an "asset management nation" strategy. Japan is looking to more actively mobilize domestic savings to fund investments and help keep government financing costs manageable. Each of these shifts could have significant long-term implications for the currency.

The new fiscal approach means keeping r below g

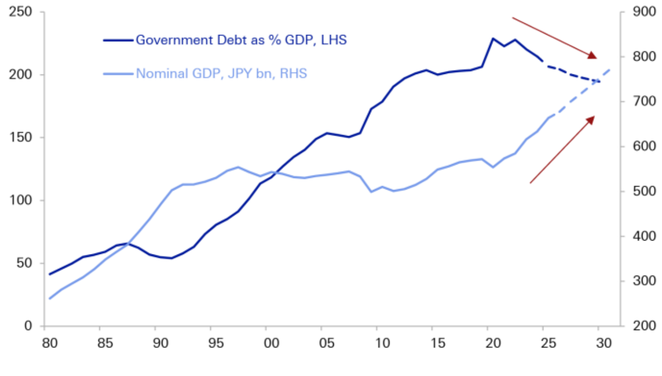
We think debt sustainability arithmetic has become key to understanding Japan's economic policy. Recall that: $\Delta d_t \approx (r-g)d_{t-1} + pd_t$ where d is the debt-to-GDP ratio, pd is the annual primary deficit, g is the nominal GDP growth rate, and r is the average interest rate paid on outstanding debt. The formula implies that Japan can run primary deficits and still achieve a declining debt-to-GDP ratio, as long as r is below g. In other words, Japan's cost of funding needs to be below nominal GDP growth for Japan to have fiscal deficit space. One might consider Abenomics' bid to return Japan to inflation as crucial to putting it on a path to fiscal health. As Figure 2 illustrates, the peaking in Japan's debt-GDP ratios has gone hand-in-hand with the revival in Japan's nominal GDP.

Figure 1: Japanese potential growth has declined on decades of underinvestment – this could now change



Source: Deutsche Bank Research, Bank of Japan

Figure 2: Rising nominal GDP has helped bring down debt-to-GDP



Source: Deutsche Bank Research, Haver Analytics

The extent of fiscal deficit space will crucially depend on the interplay between r and g with Figure 3 illustrating the potential space created by different (r-g) scenarios. We note that the Honebuto outlined a 3% nominal GDP target (g) for Japan till 2040, made up of a 2% inflation target, and 1% real growth target, with aims of increasing it. This perhaps sheds light on why the government has become sensitive to the 3% level on 10Y yields. To be sure, Japan's overall cost of funding remains much lower for now at near 1% given just a fraction of debt is refinanced every year, but there are points of vulnerability. While the weighted average maturity of debt outstanding is around 9 years, BoJ holds close to 50% of outstanding JGBs against which they pay interest on excess reserves to banks. This means that almost half the consolidated liability base is in fact linked to front-end floating rates. Therefore, not only are long-run government bond yields above 3% not likely to be sustainable, front-end rate hikes also contribute to consolidated fiscal costs. Very roughly, every 100bps of rate hikes costs Y5tn or around 0.7% of GDP.

The (r-g) formula not only explains the importance of understanding the drivers of 'r', but also explains the sensitivity to 'g'. It helps one appreciate why policymakers remain so worried about upsetting the balance on hard-won nominal GDP growth. Indeed, Japan tends to react very cautiously and dovishly to any signs of growth risks (tariffs, Iran war) and has preferred to let inflation run hot and durable before tightening.

Japanese policy makers face a fine balancing act in optimizing this formula: dovish monetary policy may help to support g, but is also driving up r via inflation expectations reflected in long-run bond yields. Conversely, hawkish monetary policy could hurt 'g' and also raise the cost of 'r' via reserve payments. In light of this, the focus on trying to manage long-end yields makes more sense.

How might Japan do this? Changes to supply-side of debt management have been tried via reduced long-end issuance; the government may now be trying to more actively influence demand.

An Asset Management Nation

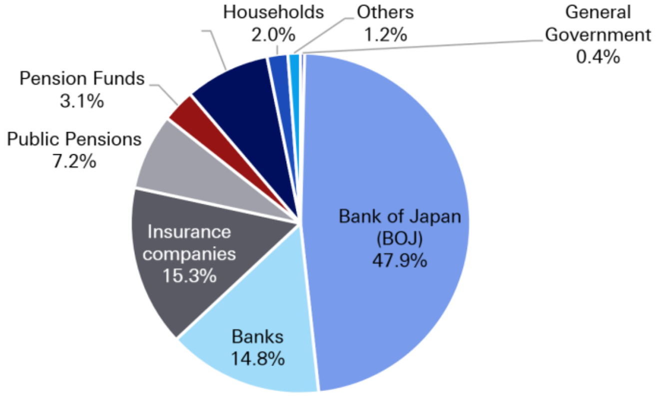
Very simply: Japan needs to find more buyers of JGBs (Figure 4). It is natural to look at where its own domestic capital has been invested. The answer points to two big places: in cash and overseas. Japan has close to USD5tn in portfolio assets held abroad across central bank, household, pension fund, insurance, and bank balance sheets, more than the size of GDP. Indeed, our research has long shown that one of the best metrics of fiscal space is a country's external position, and Japan may need to draw on it. Intuitively, the bid for strategic autonomy around the world should coincide with a growing "nationalism of capital." If industry, energy, defence all need to come home, so might the capital to fund it. We have explored this in earlier research here. More of Japan's money could now be needed at home. Finance Minister Katayama has been explicit that Japan's ambitions to create a growth-oriented economy will coincide with a "strong stock market and positive interest rates" in which domestic citizens should reap benefits. The other perspective is they will help provide the money.

Figure 3: Keep g above r is key to Japan's fiscal space. Japan does not want to return to the lost decades

Scenario	Nominal GDP	Nominal Rates	Debt Stabilizing Primary Balance	Scenario Description
1	4.5%	1.75%	**-5.5%**	*2025 Regime*
2	3.0%	2.0%	**-2.2%**	*Japan manages rise in yields to keep them below g*
3	0.0%	1.5%	**1.5%**	*Return to Lost Decades (1995-2010)*

Source: Deutsche Bank Research, Bloomberg Finance LP

Figure 4: Japan will need to find more buyers of JGBs



Source: Deutsche Bank Research, Ministry of Finance

Where the money could come from

Pension fund and household assets appear to be the two big areas of focus for markets. The government pension fund, GPIF, currently has 50% of its \$1.8tn portfolio abroad. Meanwhile, households have around 50% of their \$15tn (!) of savings in cash and deposits (Figure 5).

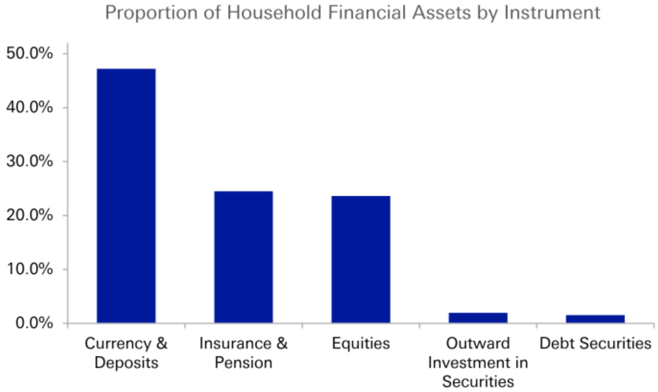
Movement of household deposits into JGBs could be encouraged through a few channels. JGBs may be brought under the NISA tax-exempt investment scheme; the retail JGB offering may be expanded or made more attractive. There were also references in the Honebuto to advancing on-chain finance, with the power of tokenization potentially being tapped to encourage financialization of savings. Mobilizing household savings could be material for yield management given their sheer size. For context, the 1000tn yen of household deposits are almost the size of the entire JGB market. It is not however likely to have any FX impact as it would mostly be an internal domestic flow.

In contrast, a shift from GPIF to invest more in domestic assets could be *very* material for FX, with PM Takaichi noting this is a key initiative. GPIF's current policy mix prescribes 25% investment in domestic equities and debt respectively, with +/-6% bands around it. If GPIF simply shifted to the top end of current permissible ranges from current levels (as of March 2026), this could bring around $200bn back to Japan across domestic equities ($130bn) and domestic bonds ($75bn).

The bigger shift to play for however is an early policy review of GPIF's overall asset mix that revises up the domestic asset targets themselves. In Figure 6 we illustrate scope for inflows at different levels of target allocation to domestic bonds alone. Hypothetically, if the target share of domestic bonds doubled to 50%, over USD400bn could flow into JGBs from GPIF. While this may seem extreme before Abenomics in 2012, GPIF had a 67% target allocation to domestic bonds in their policy mix. One could reasonably expect any flows to occur with a glide path, but the scale alone could be significant enough to impact the currency.

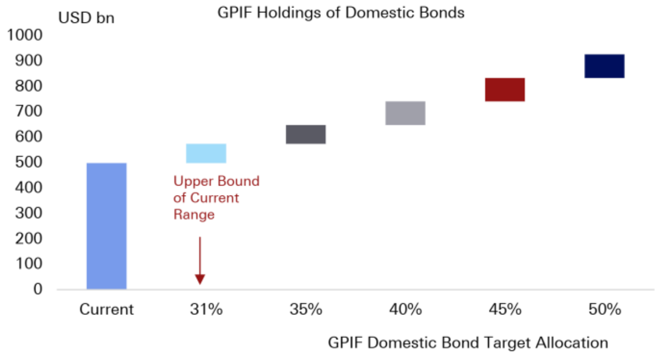
We note that GPIF has a current return objective of a 1.9% real return above nominal wage growth. The extent of any domestic reallocation will depend on how much Japan's domestic assets can deliver that return, or whether government pension returns are made subservient to financing costs for now. Another group that may be called upon to play a bond absorption role could be local banks. This is a source of financial repression has been used extensively around the world and could return where deficit financing demands it.

Figure 5: Japanese households hold half their savings in cash and deposits



Source: Deutsche Bank Research, Haver Analytics

Figure 6: GPIF could be encouraged to bring more money back into domestic bonds



Source: Deutsche Bank Research, GPIF

A shift to yield management could lead to bigger FX moves

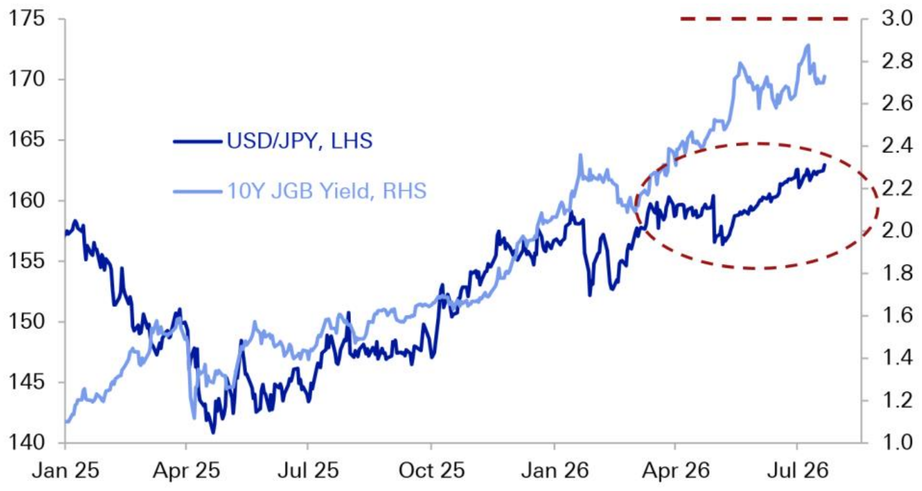
Over the past few months, Japanese policy makers have arguably been solving for stabilizing the currency. Household angst and imported inflation risks have likely encouraged this. Policymakers have engaged in verbal, rate check and actual intervention this year. And, Japan has in fact been successful in keeping USD/JPY near 160 amid the US-Iran war, a major global energy shock, and volatility in Fed pricing.

If, however, fiscal capacity is becoming the biggest and more existential policy criteria for Japan, then incentives may be shifting from FX management to yield management: from capping USD/JPY to capping 10Y yields and borrowing costs.

Suppressing interest rate volatility may come at the cost of amplifying it in FX, depending on the specific levers that the government chooses to pull. If GPIF is mandated to bring a larger share of their assets back into domestic assets, this would be very bullish for the JPY. We have previously identified GPIF as the biggest weapon in Japan's external arsenal. Equally, if BoJ is encouraged to accelerate tightening to manage inflation risk premium in long-end yields this too could be bullish JPY. If, however, the BoJ is coopted to support yield management through a more active role in the bond market, potentially resuming JGB purchases, this would be negative for the JPY. Similarly, if BoJ is encouraged to keep monetary policy dovish to prioritize (r-g) considerations this too would be bearish JPY.

Ultimately, the direction of travel for USD/JPY will depend on which measures the government favours. Clearer and more imminent signs of GPIF repatriation would favour the stronger yen tail, while pressure on BoJ to manage rates and yields again could open the weaker tail. Efforts to suppress yield volatility could come alongside bigger moves in FX from here. This favours higher JPY vols. We note USD/JPY vols are near multi-year lows with a 1Y USD/JPY straddle breaking even roughly below 150 or around 170.

Figure 7: Policy focus could be shifting from capping FX to capping yields



Source: Deutsche Bank Research, Bloomberg Finance LP

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